A Theory of Land Finance and Investment-Led Growth

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Abstract

Managing land has been a challenge for governments throughout history. I develop a growth model in which the government finances public investment with land sale and rent. A benevolent government optimally front-loads investment but maintains a *constant stock* of land supplied to the private sector, balancing fundraising through controlled supply against the welfare loss from reduced land utilization. This optimal allocation can be implemented on a balanced budget by adjusting the mix of land sale and rent, or in a time-consistent manner by issuing a full set of bonds across the maturity structure. A discretionary government on a balanced budget, however, would gradually increase land supply, consistent with historical patterns. Finally, I show that a contingent land contract linking land supply to the provision of public capital can restore the optimal allocation.

Keywords: neoclassical growth, public investment, optimal fiscal policy, financial constraints, time inconsistency, land contracts, Coase conjecture.

JEL: E62, G10, H27, H54, H63, L12, O23, O40, Q24

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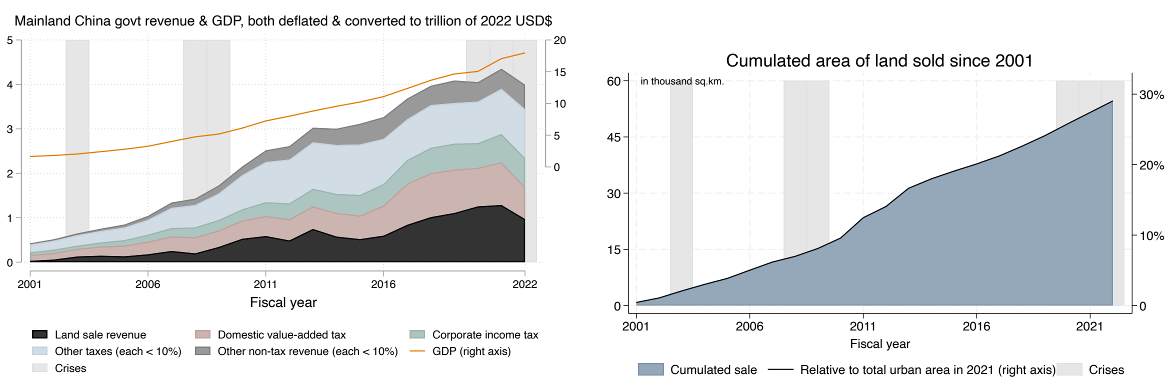
1 Introduction

Throughout history, land has been one of the most important assets for governments. Various governments—including Singapore, Hong Kong, mainland China, and the 19th-century US—have engaged in large-scale land transfers to the private sector to fund public infrastructure investment during periods of rapid economic growth, a practice known as "land finance." This practice is gaining further traction across the developing world (Peterson, 2008; UK FCDO, 2015). In the case of mainland China, Figure 1 shows that the government derives about 1 trillion US dollars of revenue from land transfers, which accounts for about a quarter of total government revenue and supports massive infrastructure investment. The government continuously expands its land supply: as of 2021, about 30% of urban land in China was only made available in the last two decades.

Figure 1: Government revenue and land supply in mainland China



The question of land management has a rich intellectual history (George, 1879), gaining renewed prominence with an open letter to Gorbachev by leading western economists amid the Soviet Union's dissolution (Tideman et al., 1990). Yet there is no formal macroeconomic model. This paper develops a dynamic general equilibrium model to study the mechanism of land finance. It formalizes certain intuitions in Tideman et al. (1990) and qualifies their proposal of government only collecting revenue from land rent. It discusses the role of commitment in light of the Coase (1972) conjecture and offers a general equilibrium resolution. The model suggests that the optimal use of land finance involves front-loading public investment while maintaining a *constant stock* of land supplied to the private sector. This optimal allocation can be implemented on a balanced budget or in a time-consistent manner. A discretionary government on a balanced budget, however, would gradually increase land supply, as observed in reality. It further proposes a welfare-improving design of land contract which links land supply to the provision of public capital.

I consider an economy where production uses public capital as an external input, provided by the government, as in Barro (1990). Unlike Barro (1990), the government finances its investment through land sale or rent rather than tax, and land is valued by households as housing consumption. I show that, in the absence of a budget constraint for the government, the *first-best* allocation in this economy features a path of economic growth exactly the same as a neoclassical growth model. Investment declines over time relative to GDP, whereas consumption-to-GDP ratio rises. As in the neoclassical growth model, the long-run outcome is independent of the initial conditions. Further, all the land should be supplied to maximize household utility.

Then I study the optimal policy of a benevolent government under a budget constraint, i.e., the *second best*, assuming complete markets and commitment. The model formalizes the positive feedback loop between economic growth and land finance articulated by Tideman et al. (1990): current land value embeds future economic growth, and a higher land value enables the government to invest more which aids economic growth. It predicts that while the government still optimally front-loads public investment, it supplies a *constant stock* of land while leaving the rest idle. The constant supply prediction follows from the government's fundraising motive and consumption-smoothing motive. Under inelastic demand for land, the benevolent government restricts land supply to raise funds for public investment while balancing this against the welfare loss of reduced land consumption. Given the need for restricting land supply, it does so in a time-invariant way to minimize the welfare loss, i.e., consumption smoothing. The optimal land supply is a multiple of the amount owned by the household, which is a quantity version of the Lerner (1934) rule obtained under inelastic demand.

Relative to the first best, the second-best interest rate is lower and household consumption grows more slowly. That happens because the government trades off the distortions incurred by restricted land supply and altered path of household consumption. Under the second best, the land income is back-loaded whereas the public investment is front-loaded. Thus, in addition to raising land income by restricting land supply, the government also suppresses the interest rate (and hence the consumption growth rate) to raise the present value of land income relative to public investment.

Moreover, the second-best allocation grows towards a steady state that depends on the initial conditions, unlike the first best. In particular, if the initial stock of public land holding or public capital is low, the long-run capital stock, output, and consumption are all lower. That is, a poorer government leads to permanently worse outcomes.

Admittedly, the assumptions of complete markets and commitment are both strong. Regarding the former, Tideman et al. (1990) suggest that the government should retain land ownership and raise revenue only from rent, concerned that large-volume land sales would depress prices and exclude credit-constrained buyers. On the latter, the Coase (1972) conjecture postulates that a durable good monopoly cannot exercise market power without commitment as it will flood the market with supply. Nonetheless, I show that the optimal allocation can be implemented without necessitating government or private sector borrowing by selling land at a particular pace. And it can be implemented in a time-consistent manner by issuing a full set of bonds across the maturity structure.

The balanced-budget implementation is possible thanks to the dual property of land as both a consumption good and an asset. My model illustrates that, since the optimal investment is front-loaded whereas the land rent income is back-loaded, without selling land, the government runs a budget deficit initially and a budget surplus after. Hence the Tideman et al. (1990) proposal is not on balanced budget. I show that the government can implement the second best on a balanced budget, by gradually selling land and lowering the amount of rental land while maintaining the total supply fixed. Essentially, selling a piece of land front-loads all future rents into current revenue. This asset property of land can be used to substitute for government borrowing to align the timing of government income and investment.

The optimal allocation can also be implemented in a time-consistent manner. This is against the backdrop of the renowned Coase (1972) conjecture that a durable good monopoly would flood the market with supply rather than restricting it. It is known that if the monopoly only rents the good out instead of selling it, they are not subject to the time consistency issue (Bulow, 1982). However, that does not implement the optimal allocation, since the rent income mismatches investment need in timing, necessitating government borrowing which introduces time consistency issue of its own. I build on the insights of Lucas and Stokey (1983) to offer a general equilibrium resolution of the Coase conjecture. In my model, the interest rate—rather than taken as given in the Coase conjecture literature—is endogenous to consumption growth, which in turn depends on the government's investment. Consequently, if there are long-term bonds outstanding, the government's investment decision would alter the interest rate and revaluate the bonds.

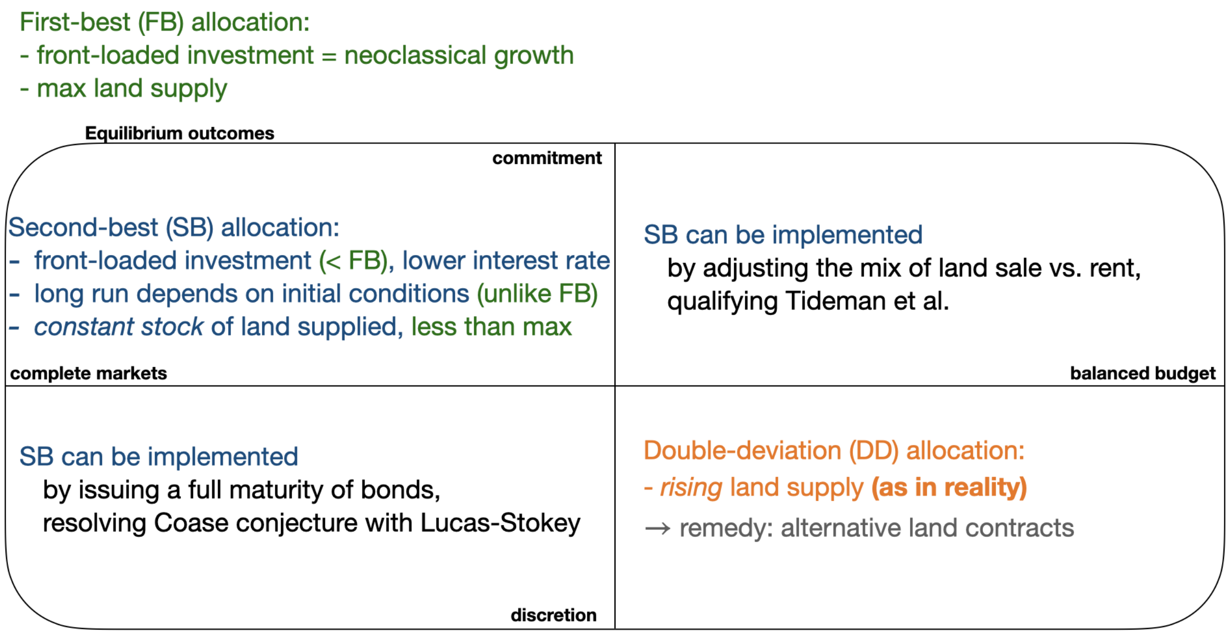
When markets are complete, the government can issue a set of bonds of different maturities up to infinity such that they would revaluate adversarially if future governments were to deviate from the optimal course of action. That is, it uses bond revaluation as a commitment device.

However, I show that when the government is simultaneously discretionary and on a balanced budget, it will still front-load investment but continuously increase land supply to the private sector. This phenomenon conforms to the Coase conjecture but in a smooth manner, as the monopoly gradually rather than immediately flood the market. The intuition is that a discretionary government always takes past sales as given and always wants to sell more to fund investment. When the capital depreciation rate is low, eventually, the government supplies all the land it has, which is distinct from the second-best long run outcome with restricted supply.

Finally, given that continual expansion of land supply indicates a deviation from the optimal allocation, I ask if an improvement is possible. I propose a novel land contract that links land supply to the provision of public capital to discipline future governments. Specifically, the optimal land contract penalizes future governments' deviation by obliging them to give more land to the contract-holders if the public capital stock is lower. This contract, whose institutional requirement is the same as administering land sale, helps illustrate the direction of policy design that can improve welfare. These theoretical predictions are summarized in Figure 2.

Literature review. This paper relates to the literature on optimal fiscal policy since Ramsey (1927), which typically assumes an exogenous stream of government expenditure other than Barro (1990) and studies taxation. I endogenize public spending and study land income via sale and rent. That literature produces important insights regarding tax smoothing (Barro, 1979, Aiyagari et al., 2002) and time consistency (Kydland and Prescott, 1977, Lucas and Stokey, 1983, Angeletos, 2002, Buera and Nicolini, 2004, Bianchi and Mendoza, 2018, Debortoli, Nunes and Yared, 2017, 2021, 2022, Jiang, Sargent and Wang, 2026), which I connect to. In the context of economic development, Acemoglu, Aghion and Zilibotti (2006), Liu (2019), and Itskhoki and Moll (2019) study optimal government policies regarding technology adoption, input-output networks, and financial frictions, and Acharya, Parlatore and Sundaresan (2025) discuss the issue of government expropriation of infrastructure funded by the private sector. This paper has a complementary focus on

Figure 2: Summary of results



land income and public investment.[1]

This paper treats the Coase (1972) conjecture in general equilibrium. Given the importance of land, it is no wonder that Coase opens his essay by considering a monopoly owning all the land in the United States. The following literature that formalizes his argument (Stokey, 1981, Bulow, 1982, Gul, Sonnenschein and Wilson, 1986), however, typically considers examples that are more microeconomic in nature, such as cars and diamonds. They posit a constant interest rate and a durable good demand curve that is independent of other consumption goods. In contrast, this paper endogenizes both the interest rate and the land demand from household optimization in general equilibrium. Because the interest rate depends on the monopoly's action, the monopoly can restrict its land supply by issuing a set of bonds across the full maturity structure, which offer a general equilibrium resolution of the Coase conjecture using Lucas and Stokey (1983).

[1] A strand of literature studies land and property taxes following George (1879). Glaeser (1996) provides a partial equilibrium model of property tax, and Oates (1969) and Jiang, Jing and Zhang (2025) study its asset pricing implications. I discuss the role of land tax in dynamic general equilibrium via a model extension: Under the second best, a positive land tax is equivalent to reducing the time-0 private land ownership, since the government captures part of its economic value. Even with a land tax, there remains an incentive to shrink land supply in order to raise revenue by increasing the total land value.

My study of land differs significantly from the study of exhaustible resources (such as oil and minerals), which was once a prominent topic of economic research (Hotelling, 1931, Nordhaus, 1973, Solow, 1974, Stiglitz, 1974, Dasgupta and Heal, 1974, Hartwick, 1977). First, land does not depreciate. Consumption smoothing of land calls for a time-invariant stock of supply, rather than a constant flow. Moreover, land is non-tradable, which means its demand is endogenous to the country's growth, whereas oil can be sold to an international market. Finally, in practice, only a few countries are endowed (or burdened) with significant exhaustible resources,[2] while managing land is a challenge and an opportunity for every nation.

Land underlies the housing market, which is important for the macroeconomy. Especially since the financial crisis, the literature has incorporated housing market into quantitative macro models (Davis and Heathcote, 2005, Favilukis, Ludvigson and Van Nieuwerburgh, 2017, Kaplan, Mitman and Violante, 2020, Greenwald and Guren, 2025), with some work in the context of China's land finance (Jiang, Miao and Zhang, 2022, Song and Xiong, 2026). They focus on demand-side credit constraints and assume simple land supply rules, whereas this paper models land supply as an optimal policy choice and explains the continuous expansion of supply in reality.[3] Given the importance of land finance for China's growth miracle and housing market (Liu, 2018, Gyourko et al., 2022), many papers have studied it empirically (Fang et al., 2016, Glaeser et al., 2017, Rogoff and Yang, 2024, He et al., 2025, Chang, Wang and Xiong, 2026). This paper illustrates the roles of financial frictions and commitment, which could be studied empirically in future work.

2 The Economy and First-Best Allocation

I first set up the economy and discuss the first-best allocation that maximizes welfare only subject to resource constraints. Under the first-best allocation, the government does not restrict land supply.

[2] Economies with abundant exhaustible resources may grow less rapidly than resource-scarce economies, known as the "resource curse" (Sachs and Warner, 1995, van der Ploeg, 2011, Venables, 2016).

[3] Stull (1974), Helpman and Pines (1977) and Rossi-Hansberg (2004) study the optimal spatial distribution of business and residential land in a city with a static model, whereas I study the amount of land supply in the whole economy in dynamic general equilibrium.

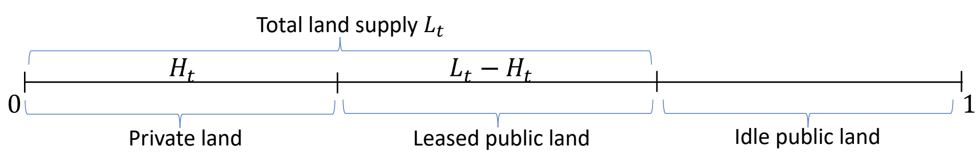
2.1 The Economy

The economy is fully deterministic. An infinitely-lived representative household derives utility from (non-durable) consumption C_t and land L_t

$$\mathcal{U}_0 \equiv \int_0^\infty e^{-\rho t} U(C_t, L_t)\, dt = \int_0^\infty e^{-\rho t} \left(\ln C_t + \nu \frac{L_t^{1-\sigma} - 1}{1-\sigma} \right) dt, \tag{1}$$

with $\rho > 0, \sigma > 1$. For simplicity, I assume that the household derives utility from land without modeling housing structure. The assumption of $\sigma > 1$ ensures that the demand for housing, holding fixed consumption C, is inelastic, as will become clear later. I normalize the total amount of land in this economy to one without loss of generality. Time-t housing consumption consists of H_t amount of owned land that is valued at price P_t, and $L_t - H_t$ amount leased from the government at rent D_t, as illustrated in Figure 3. Allowing the government to both sell and rent captures various forms of land contracts in reality such as sale, long-term and short-term leases. At time t, the household could also buy or sell housing at rate $\dot{H}_t$ at the prevailing price P_t which will determine the owned amount H_{t+dt} in the next instance. Land does not depreciate.

Figure 3: Land utilization in the model



The household supplies one fixed unit of labor. The production uses public capital stock Z_t as an external input with total factor productivity (TFP) A.[4] The stock Z_t depreciates at rate $\delta > 0$. For concreteness, I may refer to the public capital as infrastructure, but one may also include other forms of capital or public service with a higher δ. For the production, the household takes Z_t as a public good, which is the only externality in this model. The household earns labor income Y_t. The infrastructure investment is thus carried out by the government. The time-t resource constraint is

$$C_t + \dot{Z}_t + \delta Z_t \leq Y_t(Z_t) = A Z_t^\alpha, \tag{2}$$

[4] Growth in TFP and population will be accommodated in an extension considered in Appendix **??**, but they do not change the prediction that the second-best allocation features *constant* amount of land supplied to the private sector.

with $\alpha \in (0, 1)$. The production function is of a neoclassical nature, with an incarnation of public investment à la Barro (1990). I denote the consumption share (consumption-to-GDP ratio) as

$$\chi_t \equiv \frac{C_t}{Y_t}$$

and thus the investment rate (investment-to-GDP ratio) is $\frac{\dot{Z}_t + \delta Z_t}{Y_t} = 1 - \chi_t$. The reality of declining government-investment-to-GDP ratio means that χ_t increases over time.

The economy starts at time 0 with public capital stock Z_0 and land in private hands of the amount $H_0 < 1$. The government has no debt when the economy starts, and is endowed with the rest of the land $(1 - H_0)$. At each time t, the amount of land the government supplies L_t and the household owns H_t satisfy

$$H_t, L_t \in [0, 1]\,. \tag{3}$$

The government derives income from land sale and rent as their only revenue source to fund government investment. The government is allowed to rebate extra money to the household in a lump-sum manner (but they may not want to), but cannot levy lump-sum tax.

Calibration.[5] $\delta = 5\%, \rho = 2\%, \alpha = 10\%$ (Bom and Ligthart, 2014), $\sigma = 2$ (which corresponds to a demand elasticity of land/housing of one half), $A = 1$. To calibrate ν, using the FOC $U_C = U_L D^{-1}$ at the steady state and $P = \rho^{-1} D$ to get

$$\nu = \rho \frac{PL}{C} L^{\sigma - 1}$$

Take $\rho = 0.02$, housing-value-to-GDP ratio of 2 and $L = 1$, we get $\nu = 0.04$.

2.2 Remarks on Model Assumptions

Before presenting results, I provide some remarks on the model. This model is designed to minimize deviation from a neoclassical growth model while offering insights into land finance. Appendix **??** offers a few formal extensions.

1. I assume away lump-sum tax and land tax as proposed by George (1879). Pure

[5] Ratner (1983), Aschauer (1989) and Munnell (1990*a*,*b*) estimate the output elasticity of public capital. See Bom and Ligthart (2014) for a review and Leduc and Wilson (2012) and Ramey (2021) for recent advances.

land tax is practically non-existent. Property tax exists, but it distorts investment in housing structure, and is hard to implement (especially for developing countries) as it requires pricing off-market houses. An insight behind the celebrated Henry George theorem is that land tax is *non-distortionary* as land is in fixed supply. In Appendix B.1, I suggest a limit of this logic: while land tax is non-distortionary, it may be *insufficient* to fund "best outcome" if the tax base (total land market value) is low. The incentive analyzed in this paper that the government may lower land supply in order to raise its market value can still play a role, in which case there will be welfare loss from reduced land utilization, even if the government can collect land tax.

2. I assume the land is exclusively designated for housing purposes. In reality, land is used for production too and is typically zoned for different uses. The insight that a less-than-maximum supply of land ($L_t < 1$) may be chosen to increase public revenue under inelastic demand even if it incurs welfare cost does not hinge on the exact use of land. Appendix **??** considers land used as factor of production instead of housing. I further assume for simplicity that building infrastructure does not use land, which is harmless if the land used for infrastructure does not exceed the amount of land optimally left idle.

3. I abstract away from various distortionary taxes for simplicity. This model could be augmented to feature endogenous labor supply and income tax. In that case, it is desirable to use both instruments (labor tax and land supply) to raise funds and balance between multiple sources of welfare loss (labor supply distortion and lowered land utilization). A further reason not to include private capital and capital tax is that the time consistency issue with capital tax is well-known. Shutting that down helps isolate the time consistency issue with land.

4. This model is interesting only when there is a non-trivial amount of land owned by the government at time 0. This is the case for the US and Hong Kong due to their colonial histories. For other economies this model can be viewed as analyzing the policy problem *after* the government acquires land from the private sector. In Singapore, the government already owned 44% of land in 1960 (before independence), and it passed the Land Acquisition Act in 1966 (after independence) to acquire more land for public purpose at a fixed price. In mainland China, the government acquires land from rural collectives (groups of local farmers) and "rezones" it as urban land.

Through acquisitions at a fixed price, a government can obtain the land, engage in infrastructure investment which raises the land value, and thus profit from the appreciation of land. We directly place ourselves in the second stage where the government already has land at its disposal.

2.3 First-Best Allocation

The first-best (FB) allocation maximizes household utility (1) subject to the resource constraint (2) and land constraints (3) only.

Proposition 1. (First-best allocation) *The FB allocation* $\left(C_t^{FB}, Z_t^{FB}, L_t^{FB}, H_t^{FB}\right)$ *is completely characterized as follows, given* Z_0*:*

1. *the consumption growth follows*

$$\frac{\dot{C}_t^{FB}}{C_t^{FB}} = Y_t'\left(Z_t^{FB}\right) - (\delta + \rho), \tag{4}$$

subject to the transversality condition and the resource constraint

$$\dot{Z}_t^{FB} = Y_t\left(Z_t^{FB}\right) - C_t^{FB} - \delta Z_t^{FB};$$

2. *the land supply is maximum*

$$L_t^{FB} = 1;$$

3. *the amount of land held in private hands* H_t^{FB} *for* $t > 0$ *can take any value in* $[0, 1]$.

The path of consumption and public capital at the first-best allocation in this economy is identical to the path of consumption and (private) capital in a neoclassical growth model. However, the market outcome in a neoclassical model coincides with the first best, whereas this economy embeds a market failure that the household takes the public capital as given, which may inhibit the FB allocation. I will show that, in order to generate more profits, the government may lower land supply. This generates a tradeoff between fundraising and welfare loss from reduced land consumption.

Under constant TFP A, the economy converges to a steady-state that features

$$Y'\left(Z_\infty^{FB}\right) = \delta + \rho \quad \Rightarrow \quad Z_\infty^{FB} = \left(\frac{A\alpha}{\delta + \rho}\right)^{\frac{1}{1-\alpha}}.$$

At the steady state, the consumption share is

$$\chi_\infty^{FB} = 1 - \frac{\delta Z_\infty^{FB}}{Y_\infty^{FB}} = \frac{\delta(1-\alpha)+\rho}{\delta+\rho}$$

and the investment rate is thus its complement $\frac{\delta\alpha}{\delta+\rho}$.

Throughout this paper, I assume that the initial capital stock Z_0 falls short of the first-best steady-state level, i.e.,

$$Z_0 < Z_\infty^{FB} = \left(\frac{A\alpha}{\delta+\rho}\right)^{\frac{1}{1-\alpha}} \tag{5}$$

so that the economy grows over time.

3 Second-Best Allocation

Now I restrict attention to allocations that can be sustained as equilibrium outcomes. That is, the government needs to fund its investment with land sale and rent. I start with household optimization which determines the equilibrium prices. Then I study the Ramsey planning problem in primal form for the government, which will give rise to the second-best (SB) allocation, under the assumptions of complete financial markets and commitment by the government.

3.1 Household Optimization and Conditions for Second Best

Under complete markets, the household borrowing is unconstrained other than by the transversality condition. Thus the household is subject to a present-value budget constraint

$$\begin{aligned} 0 &= \int_0^\infty Q_t\left[C_t - Y_t(Z_t) + P_t\dot{H}_t + D_t(L_t-H_t) - T_t\right]\mathrm{d}t \\ &= \int_0^\infty \left\{Q_t\left[C_t - Y_t(Z_t) + D_t(L_t-H_t) - T_t\right] - H_t\frac{\mathrm{d}(P_tQ_t)}{\mathrm{d}t}\right\}\mathrm{d}t - P_0H_0 \end{aligned} \tag{6}$$

in which Q_t denotes the bond price under normalization $Q_0 = 1$, P_t is the land price, D_t is the land rent, and $T_t \geq 0$ is the lump-sum transfer from the government.

Equilibrium prices. The household optimality conditions pin down bond price and land rent as

$$Q_t = \mathrm{e}^{-\rho t} U_{C,0}^{-1} U_{C,t} \tag{7}$$

$$D_t = U_{C,t}^{-1} U_{L,t} \tag{8}$$

and thus the land price satisfies

$$P_t = Q_t^{-1} \int_t^{\infty} Q_s D_s \mathrm{d}s = U_{C,t}^{-1} \mathcal{P}_t \tag{9}$$

in which $\mathcal{P}_t \equiv \int_t^{\infty} \mathrm{e}^{-\rho(s-t)} U_{L,s} \mathrm{d}s$ incorporates the discounted sum of marginal utility from housing. Land rent D_t declines in current supply L_t, due to diminishing marginal housing consumption utility. Land price P_t decreases in future supply $L_{\tau \geq t}$ too, since land does not depreciate. Thus the government is a durable good monopolist à la Coase (1972) and faces a time consistency issue: in order to raise current price P_t, the government wants to hold future supply low; but when future comes, there is an incentive to deviate since what is sold is sold. In this section, I assume that the government can commit, which I relax later.

Feedback between growth and land finance. Holding fixed the path of land supply $\{L_t\}$, land price increases in consumption C_t: as the economy grows but land is ultimately in fixed supply, land price appreciates. Having fiscal income derived from land allows the government to invest in public capital Z_t and helps the economy grow. Hence, there is a positive feedback between land finance and economic growth the government can activate, capturing the intuition from Tideman et al. (1990).

Fiscal requirement of FB. Now I clarify assumptions such that the first best cannot be achieved as market outcome, and that the government may profit from restricting land supply.

I assess the fiscal position evaluated at the market price determined by household optimization (7-9) under the FB allocation. The present-value of time-t government investment in infrastructure, multiplied by $U_{C,0}$ to transform into unit of utility, is

$$INV_t^{FB} = Q_t \left(\dot{Z}_t^{FB} + \delta Z_t^{FB} \right) U_{C,0} = \mathrm{e}^{-\rho t} U_{C,t} \left(\dot{Z}_t^{FB} + \delta Z_t^{FB} \right) = \mathrm{e}^{-\rho t} \left(\frac{1}{\chi_t^{FB}} - 1 \right),$$

where the last step uses the resource constraint (2) and the consumption share $\chi_t^{FB} = \frac{C_t^{FB}}{Y_t^{FB}}$. The present-value util-unit of total government spending is thus

$$INV^{FB} = \int_0^\infty INV_t^{FB} \mathrm{d}t = INV^{FB}(Z_0),$$

which implicitly depends on the initial infrastructure stock Z_0.

Holding $H_t = H_0$ fixed,[6] the government derives income from land lease at each time t whose present value, again multiplied by $U_{C,0}$ to transform into util, is

$$INC_t^{FB} = Q_t D_t (1 - H_0) U_{C,0} = \mathrm{e}^{-\rho t} \nu (1 - H_0),$$

and the present-value util-unit of total government income is

$$INC^{FB} = \int_0^\infty INC_t^{FB} \mathrm{d}t = \frac{\nu (1 - H_0)}{\rho}$$

Proposition 2. *The first best cannot be achieved if and only if*

$$\nu (1 - H_0) < \rho \int_0^\infty \mathrm{e}^{-\rho t} \left(\frac{1}{\chi_t^{FB}} - 1 \right) \mathrm{d}t \tag{10}$$

which holds when initial H_0 is high or initial infrastructure Z_0 is low. A sufficient condition for that is

$$\nu < \frac{\alpha \delta}{(1 - \alpha) \delta + \rho} \tag{11}$$

If the government cannot get enough income to fulfill the required investment under first best by supplying all the land to the private sector ($L_t = 1$), then the first best is not achieveable, formalized by (10). On the left-hand side, $1 - H_0$ is the amount of land government sells or rents to the household, and ν is the total market value of land in unit of utility. The right-hand side implicitly depends on model parameters that are unrelated to land (TFP A, time preference ρ, output elasticity α, initial capital stock Z_0 and depreciation rate δ).

A sufficient condition that rules out the first best is (11). It means that, even if the economy starts with the most favorable condition (i.e., $Z_0 = Z_\infty^{FB}, H_0 = 0$), the government cannot derive enough land income to replenish the depreciated capital.

[6] In fact, the claim that follows holds even if H_t varies over time, because the housing price is "fair" in incorporating all rental revenue when household borrowing is unconstrained.

We henceforth assume that (11) holds, which is true under our calibration. In this case, the welfare obtained under second best is strictly lower than the first best.

3.2 Ramsey Planning under Commitment and Complete Markets

Using the household optimality conditions (7-9) to substitute prices in the budget constraint (6), as is a standard technique in the optimal fiscal policy literature, I get the following implementability condition

$$0 = \int_0^\infty \mathrm{e}^{-\rho t} \left[\left(C_t - Y_t\left(Z_t\right) - T_t \right) U_{C,t} + \left(L_t - H_0 \right) U_{L,t} \right] \mathrm{d}t. \tag{12}$$

The absence of P_t and H_t in this condition results from the "fair price" of housing as the household faces no borrowing constraints. The discounted sum of payment to housing consumption L_t is the same, regardless of how much the household buys or rents. When both the implementability constraint (12) and the resource constraint (2) hold, the government's budget constraint holds as a result of the Walras' law.

Definition 1. *(Ramsey problem under commitment and complete markets)* A Ramsey planner maximizes the household's life-time utility (1), subject to the resource constraint (2), land constraint (3), as well as the implementability constraint (12).

I attach multiplier $\mu_t^* \mathrm{e}^{-\rho t}$ to the time-t resource constraint (2), $\zeta_t^* \mathrm{e}^{-\rho t}$ to the time-t land constraint (3), and multiplier λ^* to the implementability constraint (12). λ^* measures the *social value of public funds*. I use superscript * to indicate the optimal allocation and policy, decided at time 0 under commitment. Formally, the Lagrangian is

$$\begin{aligned}\mathcal{L}^* = \max_{\{C_{t\geq 0}, L_{t\geq 0}, T_{t\geq 0}, H_{t>0}, Z_{t>0}\}} & \int_0^\infty \mathrm{e}^{-\rho t} U\left(C_t, L_t\right) \mathrm{d}t + \int_0^\infty \mathrm{e}^{-\rho t} \left\{ \mu_t^* \left[Y_t\left(Z_t\right) - C_t - \left(\delta + \rho\right) Z_t \right] + \dot{\mu}_t^* Z_t \right\} \mathrm{d}t + \mu_0^* Z_0 \\ & + \int_0^\infty \mathrm{e}^{-\rho t} \zeta_t^* \left(1 - L_t\right) \mathrm{d}t + \lambda^* \int_0^\infty \mathrm{e}^{-\rho t} \left[\left(C_t - Y_t\left(Z_t\right) - T_t \right) U_{C,t} + \left(L_t - H_0 \right) U_{L,t} \right] \mathrm{d}t\end{aligned} \tag{13}$$

where I have used the present-value sum version of capital stock in place of its flow version for a more intuitive representation.[7]

[7] See derivation of (C1) in Appendix C.1.

3.3 Land Supply and Quantity Lerner Rule

First observe that H_t appears in neither the objective (1) nor the constraints (2, 3, 12). Thus, given the total supply $\left\{L_t^*\right\}$, the division between sale and rent is indeterminate under SB, formalized in Lemma 1. The reason behind is that as households optimize freely intertemporally, the housing price P_t depends on the total supply L_t but not on the owned quantity H_t and the household is willing to hold any quantity $H_t \in [0,1]$ at the market price. However, the level of H_t will matter for the amount of borrowing at each point in time between the household and the government, as sale can substitute for borrowing in raising fund for the government.

Lemma 1. (Indeterminate land sale under SB) *The optimal policy may feature any $H_t \in [0,1]$.*

Quantity Lerner Rule under Inelastic Demand.

Second-Best Land Supply The FOC of second-best Lagrangian (13) w.r.t. total land supply L_t is

$$0 = \mathrm{e}^{\rho t}\frac{\partial \mathcal{L}^*}{\partial L_t} = (1+\lambda^*)\, U_{L,t} + \lambda^*\,(L_t - H_0)\, U_{LL,t} - \zeta_t^*$$

where $\zeta_t^* > 0$ if and only if $L_t = 1$.

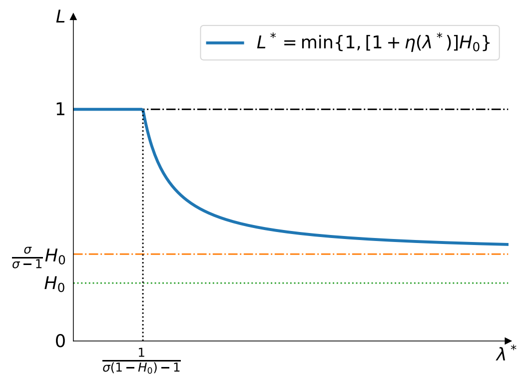
Accordingly, we can characterize the optimal land supply which equals $L^* = \min\left\{1, L^*_{interior}\right\}$ with

$$\frac{L^*_{interior} - H_0}{L^*_{interior}} = \left[1 + (\lambda^*)^{-1}\right]\sigma^{-1}. \tag{14}$$

$L^*_{interior}$ decreases in both the social value of public funds λ^* and σ, depicted in Figure 4. Comparing the SB land supply (14) to the profit-maximizing land supply (B6), the former converges to the latter when $\lambda^* \to \infty$. In that case, an additional dollar to the government leads to an infinite increase of social welfare and thus the benevolent government acts as if it maximizes profits. In the other extreme, if $\lambda^* \to 0$, it is guaranteed that $L^* = 1$: if a transfer to the government does not raise social welfare, the government should not make money by restricting land supply that hurts housing consumption.

Indeed, there exists a threshold $\bar{\lambda} \equiv \frac{1}{\sigma(1-H_0)-1}$ which is positive such that the optimal land supply is interior if and only if $\lambda^* > \bar{\lambda}$. When the marginal welfare of an extra dollar to the government is small ($\lambda^* < \bar{\lambda}$), the government prefers not to distort the land consumption. We could see this by evaluating the FOC at $L_t = 1$ to get $\mathrm{e}^{\rho t}\frac{\partial \mathcal{L}_0^*}{\partial L_t}\Big|_{L_t=1} = (1+\lambda^*)\, U_{L,t} + \lambda^*\,(1 - H_0)\, U_{LL,t}$ which is positive if λ^* is small. When the marginal welfare improvement λ^* of government making money is small, the government avoids welfare loss from reduction in housing consumption $U_{L,t}$.

Figure 4: Optimal land supply under second best



Lemma 2. (Land supply under SB) *The optimal land supply is time-invariant and declining in* λ^*

$$L^* = \begin{cases} 1, & \lambda^* \leq \bar{\lambda} = \frac{1}{\sigma(1-H_0)-1}, \\ \left[1+\eta\left(\lambda^*\right)\right] H_0 < 1, & \lambda^* > \bar{\lambda}, \end{cases}$$

with $\eta\left(\lambda^*\right) = \left(\frac{\lambda^*\sigma}{1+\lambda^*} - 1\right)^{-1}$ *denoting the supply in excess of initial household-owned land* H_0.

At the interior solution, the optimal land supply is time-invariant. That is because the utility $U(C, L)$ is separable and hence the convergence of consumption C_t does not alter the marginal utility from housing $U_{L,t}$. It is thus optimal to smooth out the housing consumption, much like the tax-smoothing result from Barro (1979).

3.4 Economic Growth

So far we have analyzed the land supply. The FOCs w.r.t. consumption C_t and public capital Z_t are

$$0 = \mathrm{e}^{\rho t}\frac{\partial \mathcal{L}^*}{\partial C_t} = \left(1+\lambda^*\right) U_{C,t} + \lambda^*\left(C_t - Y_t\right) U_{CC,t} - \mu_t^* = U_{C,t} - \lambda^* Y_t U_{CC,t} - \mu_t^*$$

$$0 = \mathrm{e}^{\rho t}\frac{\partial \mathcal{L}^*}{\partial Z_t} = \mu_t^*\left[Y_t' - (\delta+\rho)\right] + \dot{\mu}_t^* - \lambda^* Y_t' U_{C,t}$$

where μ_t^* is the resource constraint multiplier and λ^* is the present-value implementability constraint multiplier. Proposition 3 completely characterizes the SB allocation, whose proof is in Appendix C.

Proposition 3. (Second-best allocation) *The SB allocation $\left(C_t^*, Z_t^*, L_t^*, H_t^*\right)$ is completely characterized as follows, given (Z_0, H_0):*

1. *the consumption growth follows*

$$\frac{\dot{C}_t^*}{C_t^*} = \left[Y_t'(Z_t^*) - (\delta + \rho)\right] - \frac{\lambda^*}{\lambda^* + \chi_t^*}\left[\chi_t^* Y_t'(Z_t^*) + \frac{\dot{\chi}_t^*}{\chi_t^*}\right], \tag{15}$$

with consumption share $\chi_t^* \equiv \frac{C_t^*}{Y_t^*}$, subject to the transversality condition and the resource constraint

$$\dot{Z}_t^* = Y_t(Z_t^*) - C_t^* - \delta Z_t^*,$$

which jointly pin down a saddle path of convergence with consumption share strictly increasing over time, i.e., $\dot{\chi}_t^* > 0$;

2. *the land supply is time-invariant and declining in λ^**

$$L^* = \begin{cases} 1, & \lambda^* \leq \bar{\lambda} = \frac{1}{\sigma(1-H_0)-1}, \\ \left[1 + \eta(\lambda^*)\right] H_0 < 1, & \lambda^* > \bar{\lambda}, \end{cases} \tag{16}$$

with excess supply function $\eta(\lambda) \equiv \left(\frac{\lambda\sigma}{1+\lambda} - 1\right)^{-1}$;

3. *the amount of land held in private hands H_t^* for $t > 0$ can take any value in $[0, 1]$;*

4. *the multiplier λ^* is positive and satisfies*

$$\nu (L^*)^{-\sigma}(L^* - H_0) = \rho \int_0^\infty \mathrm{e}^{-\rho t}\left(\frac{1}{\chi_t^*} - 1\right)\mathrm{d}t \tag{17}$$

whose LHS simplifies to $\nu H_0^{1-\sigma}\frac{\eta(\lambda^*)}{\left[1+\eta(\lambda^*)\right]^\sigma}$ and is increasing in λ^* if $\lambda^* > \frac{1}{\sigma(1-H_0)-1}$.

The consumption growth process (15) coupled with the resource constraint (2) is a family of ordinary differential equations (ODEs) parameterized by $\lambda^* > 0$ that meets the boundary condition Z_0 and the transversality condition. The limit of $\lambda^* \to 0$ describes the

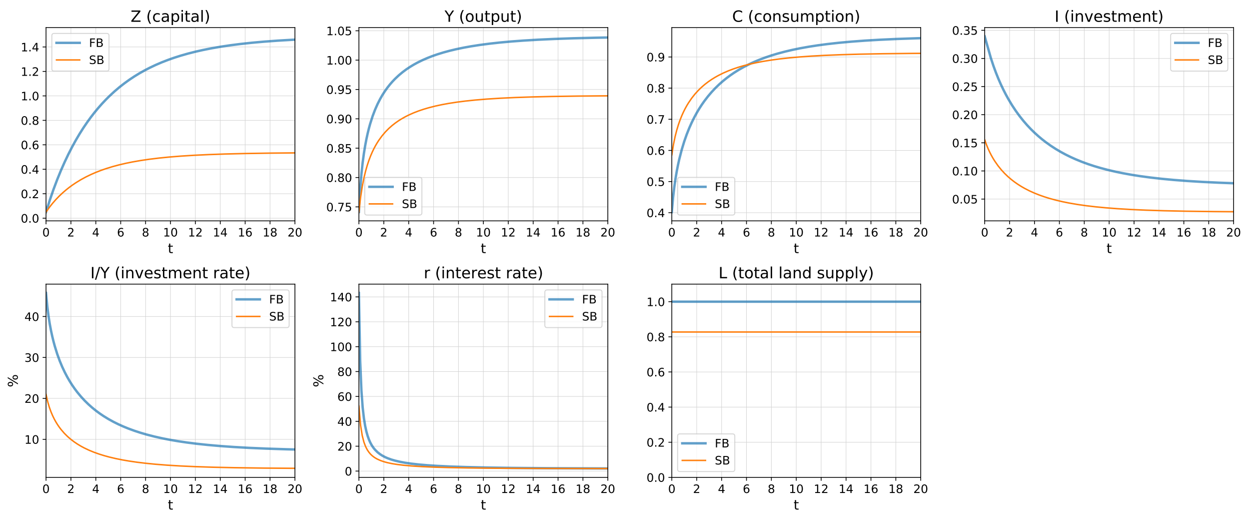
FB allocation (Proposition 1). The time-t interest rate in this economy is

$$r_t = -\frac{\dot{Q}_t}{Q_t} = \rho + \frac{\dot{C}_t}{C_t} = Y_t'(Z_t^*) - \delta - \frac{\lambda^*}{\lambda^* + \chi_t^*}\left[\chi_t^* Y_t'(Z_t^*) + \frac{\dot{\chi}_t^*}{\chi_t^*}\right]$$

Given that $\dot{\chi}_t > 0$, the SB interest rate is depressed relative to the FB interest rate $Y' - \delta$. The government is incentivized to depress the interest rate because the public investment is front-loaded but the land income is back-loaded.

Further, extending a result from Barro and Sala-i-Martin (2004, Chapter 2.6), I show that χ_t^* strictly increases during convergence. Barro and Sala-i-Martin give conditions for monotonicity of χ_t^{FB} during transition under efficient allocation, i.e. the ODE system (15, 2) with $\lambda^* = 0$. I generalize their result to $\lambda^* \geq 0$. Even when the government cannot achieve the first best, it still chooses to front-load investment (investment rate $1 - \chi_t^*$ strictly decreasing over time as the economy grows).[8]

Figure 5: Economic growth under first best (FB) and second best (SB)



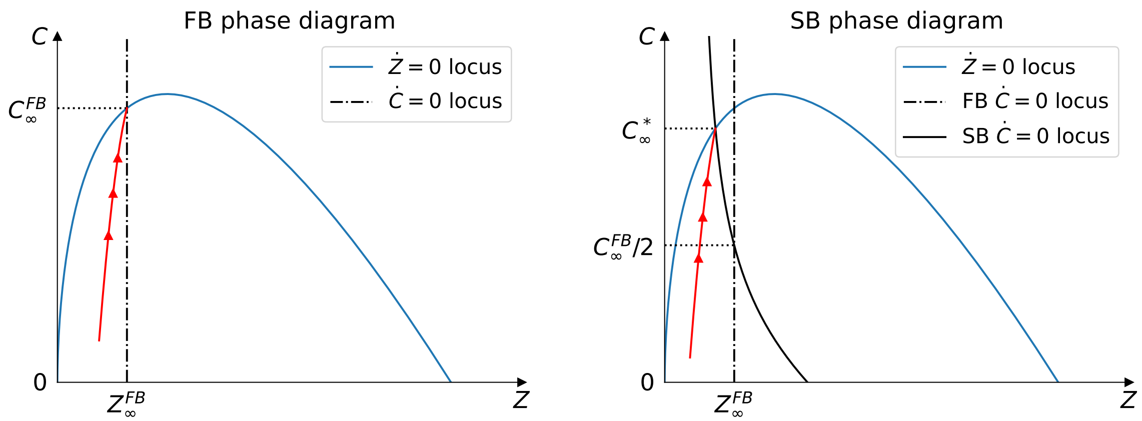
Based on (15, 2), Figure 6 plots the phase diagram in the (Z, C) space. The shape of $\dot{Z} = 0$ is independent of λ^*, as it arises from the resource constraint. When $\lambda^* = 0$, the $\dot{C} = 0$ locus is a vertical line, as in the neoclassical growth model. Denote the intersection of

[8] I assumed log utility in my model for simplicity. This functional form can be relaxed without disturbing the monotonicity of χ_t^*, as long as the elasticity of intertemporal substitution (EIS) is lower than $1/\alpha$. This inequality holds with a large margin: Havranek et al. (2015) reviews a vast literature that estimates EIS which mostly produce estimates less than or around one; Bom and Ligthart (2014) survey estimates of output elasticity of public capital α and suggest an average estimate of 0.106.

these two loci as $(Z_\infty^{FB}, C_\infty^{FB})$. When $\lambda^* > 0$, the $\dot{C} = 0$ locus rotates counterclockwise around exactly $(Z_\infty^{FB}, C_\infty^{FB}/2)$ and its interaction with the $\dot{Z} = 0$ locus shifts to the left. Indeed, I establish the following comparative statics for the steady state value.

Figure 6: Phase diagram

(a) First best (b) Second best



Proposition 4. *(Long-run comparative statics of second best)*

1. *The steady-state consumption share χ_∞^* follows*

$$\chi_\infty^*(\lambda^*) = \frac{(1-\lambda^*) + \sqrt{(1+\lambda^*)^2 + \frac{4\delta\alpha}{\delta(1-\alpha)+\rho}\lambda^*}}{2} \frac{\delta(1-\alpha)+\rho}{\delta+\rho} \in [\chi_\infty^{FB}, 1) \tag{18}$$

   which equals 1 if $\delta = 0$ and strictly increases in λ^* if $\delta > 0$. The steady-state infrastructure and consumption satisfy

$$Z_\infty^*(\lambda^*) = \begin{cases} \left[\frac{\alpha}{(1+\lambda^*)\rho}\right]^{\frac{1}{1-\alpha}} & \delta = 0 \\ \left[\frac{A(1-\chi_\infty^*)}{\delta}\right]^{\frac{1}{1-\alpha}} & \delta > 0 \end{cases} \tag{19}$$

$$C_\infty^*(\lambda^*) = \begin{cases} A\left[\frac{\alpha}{(1+\lambda^*)\rho}\right]^{\frac{\alpha}{1-\alpha}} & \delta = 0 \\ A\chi_\infty^*\left[\frac{A(1-\chi_\infty^*)}{\delta}\right]^{\frac{\alpha}{1-\alpha}} & \delta > 0 \end{cases} \tag{20}$$

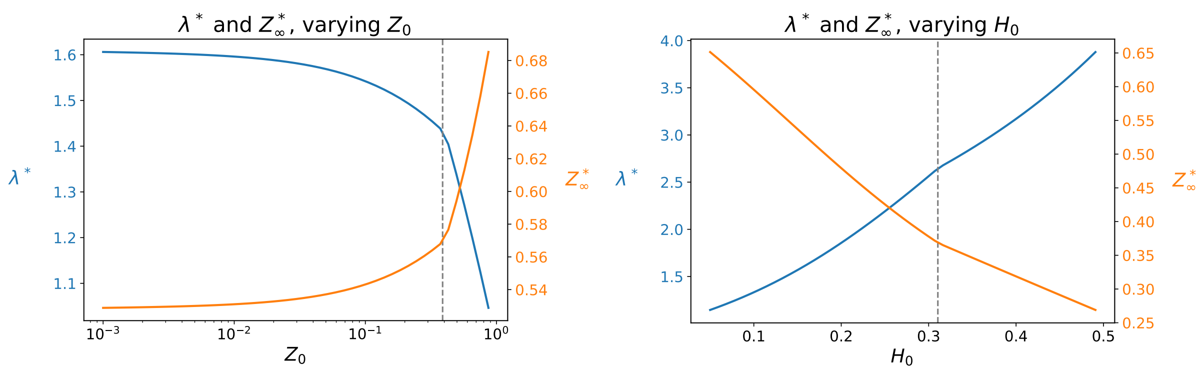
   both of which decrease in λ^* under any δ.

2. *If the initial infrastructure stock Z_0 is lower or private land holding H_0 is higher, then the multiplier λ^* is higher, the long-run consumption and infrastructure stock C^*_∞, Z^*_∞ are lower, and the long-run consumption share $\chi^*_\infty \equiv C^*_\infty/Y^*_\infty$ is higher. That is, a poorer government leads to a permanently lower GDP.*

When the marginal value of a dollar to the government is higher (higher λ^*), the economy converges to a steady state with lower consumption and lower infrastructure stock but higher consumption-to-output rate. Obviously, the steady state consumption and infrastructure are both lower under SB than under FB, and the consumption share is higher, since the FB allocation can be viewed as the limit of $\lambda^* \to 0$.

Figure 7: Long-run comparative statics

(a) Varying initial capital stock Z_0 (b) Varying initial private land holding H_0



4 Constrained Implementations of Second Best

Previously I allow the government to flexibly choose between land sale or lease, to borrow, and to commit to its future actions. Here I analyze what role each of these assumptions play. To illustrate these, I assume that the initial conditions (Z_0, H_0) are such that the second-best land supply L^* is interior for simplicity, but the results can be extended more generally.

The assumption of commitment directly circumvents the time inconsistency issue from Coase (1972). As for forms of land contracts and financial markets, recall that the investment-to-GDP ratio $1 - \chi^*_t$ strictly declines over time. Using (8), the land-rent-to-GDP ratio $\frac{D^*_t}{Y^*_t} = U^*_L \frac{C^*_t}{Y^*_t} = U^*_L \chi^*_t$ strictly declines. Thus the investment is front-loaded but the fiscal income from lease is back-loaded. Though the total fiscal income and expenditure are equal in terms of present value, the timing mismatch suggests that the government's lease income is less than expenditure early on. Hence the government has to transfer funds intertemporally. This can be done by borrowing (in which case having one bond is enough), or by selling rather than leasing land (which amounts to collecting all future rents from a piece of land today).

One may wonder if the optimal policy is changed once one of these assumptions is removed. I establish that land contract forms, financial markets, and the ability to commit, individually, are not essential for implementing the SB allocation.

4.1 Implementation with a Single Land Contract

The SB allocation can be implemented with a single land contract, i.e. either sale or lease. This is an immediate implication of Lemma 1 and Proposition 3, since H_t is indeterminate. In fact, it can be implemented by any form of land contract, as it will be priced in the financial markets. If the government sells L^* amount of land directly to the household at time 0, this maximizes the debt owed by the household and the government can save the proceeds. If the government keeps leasing L^* amount of land, it has to borrow against its future lease income.

An important piece of policy advice from Tideman et al. (1990) is that, if there are potential borrowing constraints faced by the household, the government could rent the land out on an annual basis. While it is intuitive that outright sale can be problematic under borrowing constraints, it takes a model to see if it is a good idea for the government to hold on to the land forever. On this proposal, the previous paragraph suggests that the government would run a fiscal deficit early on since the lease income is insufficient to fund expenditure. A slightest friction or reason against government deficit or borrowing could render this undesirable. Next, I show that there is a way to implement the SB allocation without necessitating government or private sector borrowing.

4.2 Balanced-Budget Implementation

I show that SB can be implemented by a government on balanced budget by adjusting the amount of land sale and rent while holding the total supply fixed.

First, we evaluate the fiscal position under SB. The time-t land rent relative to GDP is,

$$\frac{D_t^*}{Y_t^*} = \frac{U_L^*/U_{C,t}^*}{Y_t^*} = U_L^* \chi_t^*,$$

which is increasing over time. That is because the household values future housing L_t more as their future consumption C_t is higher relative to future GDP Y_t. If the government derives rental income at each time t without selling any land (i.e., $H_t = H_0$), the fiscal income is back-loaded.

In contrast, the time-t expenditure relative to GDP is

$$\frac{\dot{Z}_t^* + \delta Z_t^*}{Y_t^*} = 1 - \chi_t^*$$

which is decreasing over time. Without selling land, the rental income and investment are equal in present value but mismatched in timing, since the consumption share χ_t^* is time-varying.

Proposition 5. (Fiscal imbalance without land sales, Tideman et al., 1990) *Under the SB allocation, if the government does not sell any land (i.e., $H_t = H_0$), it runs a budget deficit initially and a budget surplus thereafter.*

Nonetheless, acknowledge that the housing price will incorporate all the future rental income and thus the government could access future rental income via land sale, a point made in Lemma 1—if the government sells $H_t = L^*$, it can extract all the land value at time 0 and roll over the savings for future expenditure. Further, as the time profiles of sale income and lease income "span" the time profile of SB expenditure, the government can tune the sale/lease ratio to exactly fund its expenditure as each time t. Proposition 6 formalizes this intuition to illustrate that borrowing is not necessary to implement SB.

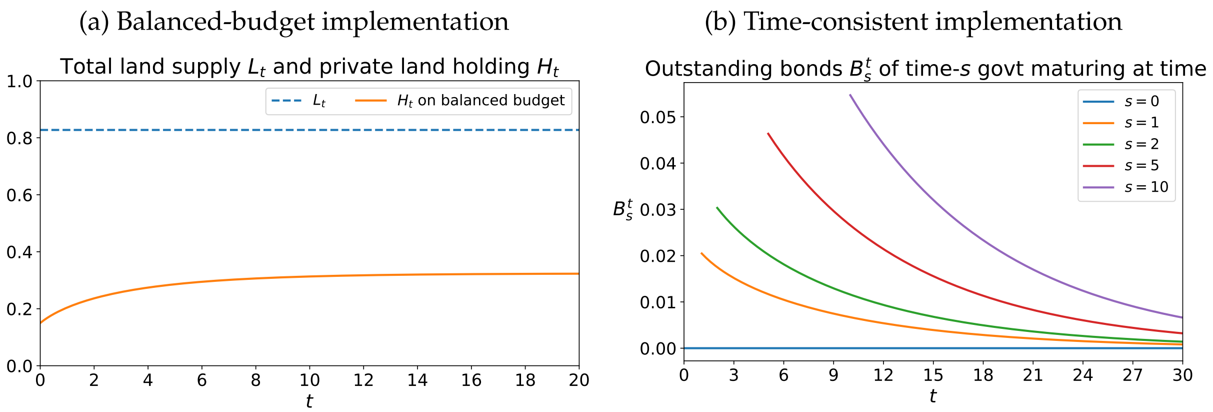
Proposition 6. *SB allocation could be achieved without borrowing by either the government or the household, if the government can commit to its future actions, by maintaining for all $s \geq 0$*

$$H_s = H_s^{NB} \equiv H_0 + \left[1 - \frac{\int_s^\infty \mathrm{e}^{-\rho(t-s)} \left(\frac{1}{\chi_t^*} - 1\right) \mathrm{d}t}{\int_0^\infty \mathrm{e}^{-\rho t} \left(\frac{1}{\chi_t^*} - 1\right) \mathrm{d}t}\right] \eta\left(\lambda^*\right) H_0, \tag{21}$$

The threshold H_s^{NB} increases in s from H_0 to $H_\infty^{NB} = H_0 + \left[1 - \frac{\frac{1}{\chi_\infty^*}-1}{\rho\int_0^\infty e^{-\rho t}\left(\frac{1}{\chi_t^*}-1\right)dt}\right]\eta(\lambda^*) H_0$ as s goes from 0 to ∞. That is, the government sells land from H_0 until H_∞^{NB}.

When (21) holds, the fiscal income exactly offsets the expenditure at any point in time, and no financial transaction is needed. As $s \to \infty$, the amount of land the government owns $L^* - H_\infty^{NB}$ generates exactly enough revenue to replenish infrastructure depreciation. If $\delta = 0$ holds in addition, and thus $\chi_\infty^* = 1$, then the government owns no land in the long run ($H_\infty^{NB} = L^*$). When (21) holds as inequality with H_s larger than H_s^{NB}, the government saves the fiscal surplus for future investment. In this case, it suffices to have only one debt instrument for the government to use.

Figure 8: Constrained implementations of second best



4.3 Time-Consistent Implementation

Last, I show that SB can be implemented in a time-consistent manner by using a full maturity of bonds as a commitment device. This builds on the insights of Lucas and Stokey (1983) and Debortoli, Nunes and Yared (2021).

Let B_s^τ be a zero-coupon bond with maturity $\tau - s$ issued at time s, which pays one dollar at time τ to the household. If the government reoptimizes at time s, taking as given $\left(Z_s, H_s, \{B_s^\tau\}_{\tau \geq s}\right)$, its implementability constraint (12) takes the following form

$$0 = \int_s^\infty e^{-\rho(t-s)} \left[\left(C_t - Y_t - B_s^t\right) U_{C,t} + (L_t - H_s)\, U_{L,t}\right] dt, \tag{22}$$

which is a discounted sum from time s to ∞. The time-s Lagrangian is

$$\begin{aligned}\mathcal{L}^{s*} = \max & \int_s^\infty \mathrm{e}^{-\rho(t-s)} U\left(C_t, L_t\right) \mathrm{d}t + \int_s^\infty \mathrm{e}^{-\rho(t-s)} \left\{\mu_t^{s*}\left[Y_t\left(Z_t\right) - C_t - \left(\delta+\rho\right) Z_t\right] + \dot{\mu}_t^{s*} Z_t\right\} \mathrm{d}t + \mu_s^{s*} Z_s \\ & + \lambda^{s*} \int_s^\infty \mathrm{e}^{-\rho(t-s)} \left[\left(C_t - Y_t\left(Z_t\right) - B_s^t\right) U_{C,t} + \left(L_t - H_s\right) U_{L,t}\right] \mathrm{d}t \\ & + \int_s^\infty \mathrm{e}^{-\rho(t-s)} \zeta_t^{s*}\left(1 - L_t\right) \mathrm{d}t + \int_s^\infty \mathrm{e}^{-\rho(t-s)} \xi_t^{s*}\left(1 - H_t\right) \mathrm{d}t\end{aligned} \tag{23}$$

in which λ^{s*} is the multiplier attached to the implementability constraint (22), μ_t^{s*} is the multiplier attached to the time-t resource constraint, and ζ_t^{s*}, ξ_t^{s*} regards the upper bound on time-t land supply. The subscript $s*$ indicates that the optimization is done at time s.

Given the initial condition $\left(Z_s, H_s, \{B_s^\tau\}_{\tau \geq s}\right)$, the time-$s$ planner solves (23) subject to the implementability constraint and resource and land constraints to arrive at the optimal allocation. The resulting multipliers $\left(\lambda^{s*}, \left\{\mu_t^{s*}\right\}_{t\geq s}, \left\{\zeta_t^{s*}\right\}_{t\geq s}\right)$ must all be non-negative. For the time-0 optimal allocation analyzed in Section 3.2 to be carried out by time-s planner, the time-0 planner solves the inverse problem and sets $\left(Z_s, H_s, \{B_s^\tau\}_{\tau\geq s}\right)_{s\geq 0}$ to incentivize future planners. If there exists a solution whose implied multipliers $\left(\lambda^{s*}, \left\{\mu_t^{s*}\right\}_{t\geq s}, \left\{\zeta_t^{s*}\right\}_{t\geq s}\right)_{s\geq 0}$ are all non-negative, then the time-0 optimal allocation can be carried out.

I characterize the set of solutions in Proposition 7, and in doing so, uncover an important property of any time-consistent implementation regarding the amount of land in private hands H_t.

Proposition 7. *SB allocation could be achieved without commitment, if the government can issue bonds in complete markets with a specific maturity structure.*

Specifically, there exists a set of implementation parameterized by $\left\{\lambda^{s*} | \lambda^{s*} \in \left(\frac{1}{\sigma-1}, \infty\right]\right\}_{s>0}$ *such that the land in private hands and the amount of* $(\tau - s)$*-maturity zero-coupon bond the government owes at time s satisfy*

$$H_s = \frac{L^*}{1+\eta\left(\lambda^{s*}\right)} = \frac{\sigma - 1 - \left(\lambda^{s*}\right)^{-1}}{\sigma} L^* \leq \frac{\sigma-1}{\sigma} L^* \tag{24}$$

$$B_s^\tau = \left(\bar{b}_s^\tau + \tilde{b}_s^\tau\right) C_\tau^*, \ \forall \tau \geq s \tag{25}$$

with

$$\bar{b}_s^\tau = \frac{C_\tau^* m_\tau^{-1}}{\int_s^\infty \mathrm{e}^{-\rho(t-s)} C_t^* m_t^{-1} \mathrm{d}t} \left(INC\left(\lambda^{s*}; L^*\right) - INV^{s*}\right) \tag{26}$$

$$\tilde{b}_s^\tau = \left(\frac{C_\tau^* m_\tau^{-1}}{\rho\int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} \mathrm{d}t} - 1\right)\left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*}\right) \tag{27}$$

In the expressions above, m_t is an integrating factor defined as $m_t \equiv e^{\int_0^t [Y'_\tau - (\delta+\rho)]\mathrm{d}\tau}$ that increases in t, $INC(\lambda^{s*}; L^*) \equiv \frac{\nu(L^*)^{1-\sigma}}{\rho}\frac{1+\lambda^{s*}}{\lambda^{s*}\sigma}$ is the discounted sum of future land income under $H_s = \frac{L^*}{1+\eta(\lambda^{s*})}$, and $INV^{s*} \equiv \int_s^\infty e^{-\rho(t-s)}\left(\frac{1}{\lambda_t^*} - 1\right)\mathrm{d}t$ is the discounted sum of future investment. Last, for any $s > 0$ and λ^{s*}, B_s^τ / C_τ^* is either a positive constant across τ or strictly monotonic (increasing or decreasing) in τ.

In (25), the $\bar{b}_s^\tau$ term indicates a baseline position whose sum across the maturity structure equals fiscal surplus (land income minus investment), and the $\tilde{b}_s^\tau$ term is a residual position that averages to zero to fine-tune the maturity structure. The baseline position $\bar{b}_s^\tau$ is positive (/negative) across τ if λ^{s*} is larger (/smaller) than a threshold $\bar{\lambda}^s$, which increases over s from value $\bar{\lambda}^0 = \lambda^*$ at $s = 0$. Further, the $\bar{b}_s^\tau$ term decreases in absolute value over τ as $C_\tau^* m_\tau^{-1}$ decreases in τ — the further into the future, the smaller bond position relative to consumption. The residual position $\tilde{b}_s^\tau$ is monotonically decreasing (/increasing) if λ^{s*} is less (/greater) than λ^*. If setting $\lambda^{s*} = \lambda^*$ (which implies $H_s = H_0$), we would exactly have $B_s^\tau = C_\tau^* \bar{b}_s^\tau = C_\tau^* \frac{C_\tau^* m_\tau^{-1}}{\int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} \mathrm{d}t}(INC(\lambda^{s*}; L^*) - INV^{s*})$ amount of government borrowing for all $\tau \geq s$. In this case, as time $s \to 0$, the bond position approaches zero since $INC(\lambda^*; L^*) = INV^*$ from Proposition 3.

Moreover, (24) implies that the land in private hands H_s never exceeds $\frac{\sigma-1}{\sigma}L^*$. To see the intuition, applying Lemma 2 to the time-s decision, the optimal land supply satisfies

$$L^{s*} = \begin{cases} 1, & \lambda^{s*} \leq \frac{1}{\sigma(1-H_s)-1}, \\ [1+\eta(\lambda^{s*})] H_s < 1, & \lambda^{s*} > \frac{1}{\sigma(1-H_s)-1}, \end{cases}$$

with a decreasing function $\eta(\lambda^{s*}) = \left(\frac{\lambda^{s*}\sigma}{1+\lambda^{s*}} - 1\right)^{-1}$. As λ^{s*} increases from 0 to ∞, L^{s*} decreases from 1 to $\frac{\sigma}{\sigma-1}H_s$. The logic behind is that, depending on the need for money, the government may supply any amount of land from $\frac{\sigma}{\sigma-1}H_s$ (maximizing profits) to 1 (maximizing housing consumption), but it will never set L^{s*} below the level $\frac{\sigma}{\sigma-1}H_s$. In order for L^{s*} to equal L^*, we must have $H_s \leq \frac{\sigma-1}{\sigma}L^*$; otherwise there is no non-negative solution λ^{s*}. That is, the time-0 planner has to leave $H_s \leq \frac{\sigma-1}{\sigma}L^*$ in private hands to incentivize future planners.

Last, this implementation requires the government to constantly adjust its bond and land positions. However, as long as the chosen sequence of multiplier λ^{s*} is continuous in

s, the adjustment is gradual rather than abrupt. One possibility is to fix $\lambda^{s*} \equiv \lambda^*$, in which case the government keeps land holdings at $H_s \equiv H_0$ and adjust bond positions $\{B_s^\tau\}$ only according to $\bar{b}_s^\tau$ since $\tilde{b}_s^\tau = 0$.

5 Double-Deviation Allocation

I have established that land contracts, complete markets, and commitment, individually, are not essential prerequisites for the implementation of SB allocation. While in reality, governments do use various land contracts (sale and lease, or long-term and short-term leases), complete markets and commitment are both unrealistically strong assumptions. Here I show that the combination of discretion and financial constraints denies SB allocation. I call this case *double deviation (DD)* for simplicity.

I assume that the government has to run a balanced budget, using income from land sale and lease to exactly pay for investment at each point in time.[9] Thus we have two state variables (Z_t, H_t).

5.1 Ramey Planning under Discretion and Balanced Budget

To tractably model discretion, I take the limit of $\Delta t \to 0$ of a discrete-time Markov Perfect Equilibrium (MPE) in which each governor is in charge for for one period with period length Δt. Each governor takes into account how her choice of (Z_t, H_t) affects the future governors' choices.

Now the budget constraint holds period by period as

$$0 = (C_t - Y_t(Z_t) - T_t)\,\Delta t + (L_t - H_t)\,D_t\Delta t + \Delta H_{t+\Delta t} P_t$$

Plugging in the discretized household optimality conditions (8, 9), I rewrite this as a period-t implementability constraint

$$0 = (C_t - Y_t(Z_t) - T_t)\,U_{C,t}\Delta t + (L_t - H_t)\,U_{L,t}\Delta t + \mathrm{e}^{-\rho\Delta t}\Delta H_{t+\Delta t}\mathcal{P}^{DD}_{L,t+\Delta t} \tag{28}$$

with $\mathcal{P}^{DD}_{L,t} \equiv \sum_{s\geq 0} \mathrm{e}^{-\rho s\Delta t} U_{L,t+s\Delta t}\Delta t$.

[9] If the government is allowed to save but not borrow, I conjecture that it is optimal not to save. The logic is that expenditure is optimally front-loaded, and if the government sells more than it needs for expenditure, this exacerbates the time inconsistency problem.

Definition 2. *(Markov perfect equilibrium in discrete time)* Each governor maximizes the household's life-time utility

$$\mathcal{V}^{DD}\left(H_t, Z_t\right) = \max_{C_t, L_t, T_t, H_{t+\Delta t}, Z_{t+\Delta t}} U\left(C_t, L_t\right)\Delta t + \mathrm{e}^{-\rho\Delta t}\mathcal{V}^{DD}\left(H_{t+\Delta t}, Z_{t+\Delta t}\right), \tag{29}$$

subject to the resource constraint (2), land constraints (3), as well as period-t implementability constraint (28).

The discrete-time Lagrangian with period length Δt is

$$\begin{aligned}\mathcal{V}^{DD}\left(H_t, Z_t\right) = \max_{C_t, L_t, H_{t+\Delta t}, Z_{t+\Delta t}} & U\left(C_t, L_t\right)\Delta t + \mathrm{e}^{-\rho\Delta t}\mathcal{V}^{DD}\left(H_{t+\Delta t}, Z_{t+\Delta t}\right) \\ & + \mu_t^{DD}\left[Y\left(Z_t\right)\Delta t + \left(1-\delta\Delta t\right)Z_t - C_t\Delta t - Z_{t+\Delta t}\right] \\ & + \lambda_t^{DD}\left[\left(C_t - Y\left(Z_t\right)\right)U_{C,t}\Delta t + \left(L_t - H_t\right)U_{L,t}\Delta t + \mathrm{e}^{-\rho\Delta t}\Delta H_{t+\Delta t}\mathcal{P}_{t+\Delta t}^{DD}\right] \\ & + \zeta_t^{DD}\Delta t\left(1-L_t\right) + \xi_t^{DD}\left(1-H_{t+\Delta t}\right)\end{aligned} \tag{30}$$

where I attach multiplier μ_t^{DD} to the resource constraint (2), λ_t^{DD} to the time-t implementability condition (28), and ζ_t^{DD}, ξ_t^{DD} to the land constraints. ζ_t^{DD} ($/\xi_t^{DD}$) is equal to zero when L_t ($/H_{t+\Delta t}$) is less than one. Differing from the SB policy analyzed in Section 3.2, here there is a multiplier λ_t^{DD}, which measures the social value of public funds for each time t, instead of a single λ^*.

Further note that $\mathcal{P}_t^{DD} = \mathcal{P}_t^{DD}\left(H_t, Z_t\right)$ is itself recursive and depends on the optimal policy rule $L^{DD}\left(H, Z\right)$, as

$$\mathcal{P}^{DD}\left(H_t, Z_t\right) = U_L\left(L^{DD}\left(H_t, Z_t\right)\right)\Delta t + \mathrm{e}^{-\rho\Delta t}\mathcal{P}^{DD}\left(H_{t+\Delta t}, Z_{t+\Delta t}\right). \tag{31}$$

This is because the time-t planner decides on $H_{t+\Delta t}, Z_{t+\Delta t}$, acknowledging that this choice affects future planners' optimal choices of $L_{t+s\Delta t}$ and thus $U_{L,t+s\Delta t}$.

5.2 Economic Growth and Land Supply

I characterize the optimality conditions w.r.t. $C_t, L_t, H_{t+\Delta t}, Z_{t+\Delta t}$ and take the limit of $\Delta t \to 0$ to arrive at the optimal equilibrium allocation in continuous time. See Appendix C for detailed derivations.

Proposition 8. (Allocation under double deviation) *The DD allocation* $\left(C_t^{DD}, Z_t^{DD}, L_t^{DD}, H_t^{DD}\right)$ *is completely characterized as follows, given* $\left(Z_0, H_0\right)$*:*

1. *the consumption growth follows*

$$\frac{\dot{C}_t^{DD}}{C_t^{DD}} = \left[Y_t'\left(Z_t^{DD}\right) - (\delta+\rho)\right] - \frac{\lambda_t^{DD}}{\lambda_t^{DD}+\chi_t^{DD}}\left[\chi_t^{DD} Y_t'\left(Z_t^{DD}\right) + \frac{\dot{\chi}_t^{DD}}{\chi_t^{DD}} - \frac{\dot{\lambda}_t^{DD}}{\lambda_t^{DD}} - \chi_t^{DD} C_t^{DD} \dot{H}_t^{DD} \left(\frac{\partial \mathcal{P}^{DD}}{\partial Z}\right)_t\right] \tag{32}$$

with consumption share $\chi_t^{DD} \equiv \frac{C_t^{DD}}{Y_t^{DD}}$ *and* $\mathcal{P}^{DD}$ *capturing present value of future marginal utility as (31), subject to the transversality condition and the resource constraint*

$$\dot{Z}_t^{DD} = Y_t\left(Z_t^{DD}\right) - C_t^{DD} - \delta Z_t^{DD};$$

2. *the land supply satisfies*

$$L_t^{DD} = \min\left\{1, \left[1+\eta\left(\lambda_t^{DD}\right)\right] H_t^{DD}\right\}, \tag{33}$$

with excess supply function $\eta(\lambda) \equiv \left(\frac{\lambda\sigma}{1+\lambda} - 1\right)^{-1}$ *which decreases in* λ*;*

3. *the amount of land held in private hands evolves according to, when* $H_t^{DD} < 1$,

$$\dot{H}_t^{DD} = \frac{\mathcal{P}_t^{DD}}{(\partial \mathcal{P}^{DD}/\partial H)_t} \frac{\dot{\lambda}_t^{DD}}{\lambda_t^{DD}} \tag{34}$$

4. *when the multiplier* λ_t^{DD} *is positive, it satisfies*

$$\nu\left(L_t^{DD}\right)^{-\sigma}\left(L_t^{DD} - H_t^{DD}\right) + \dot{H}_t^{DD}\mathcal{P}_t^{DD} = \frac{1}{\chi_t^{DD}} - 1 \tag{35}$$

whose LHS is increasing in λ_t^{DD} *holding* H_t^{DD} *fixed, when* $L_t^{DD} < 1$.

We notice a few interesting differences between the DD allocation in Proposition 8 to the SB allocation in Proposition 3. In terms of consumption growth (32), in addition to the wedge between $Y'\left(Z_t^{DD}\right) - (\delta+\rho)$ and $\dot{C}_t^{DD}/C_t^{DD}$ caused by non-zero λ_t^{DD}, which exists in the SB allocation too, another wedge emerges due to time-varying λ_t^{DD} and H_t^{DD}. In terms of total land supply (33), time-t government always supplies an extra amount relative to H_t^{DD}, which is what is already sold by time t, rather than H_0. And the added supply $\eta\left(\lambda_t^{DD}\right)$ changes over time too, as the social value of public funds λ_t^{DD} varies.

Further, I show in Proposition 9 that when the depreciation rate δ is low, the land market will eventually be saturated ($L_\infty^{DD} = 1$). If the economy grows but the land market is never saturated, assuming that the consumption share χ_t^{DD} is eventually monotonic and a few other regularity conditions, I establish in Proposition 10 that both H_t^{DD} and L_t^{DD} will strictly increase during convergence.

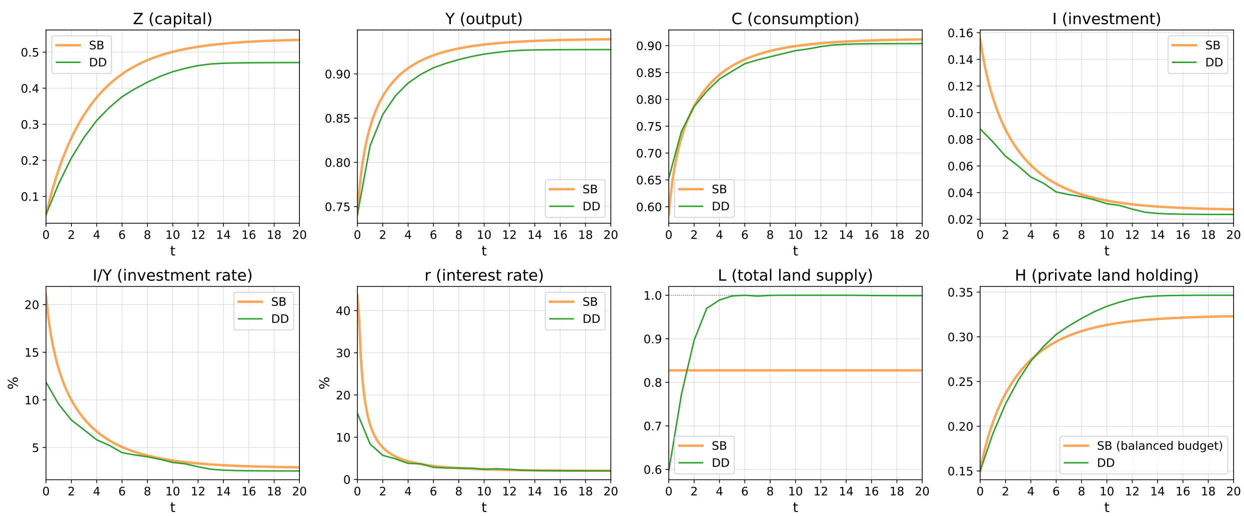
Proposition 9. *If* $\frac{\alpha\delta}{(1-\alpha)\delta+\rho} \leq 4\frac{\nu}{\sigma}\left(1+\frac{\nu}{\sigma}\right)$*, then:*

1. *The steady state is characterized by* $L_\infty^{DD} = 1$*: land supply is saturated in the long run.*

2. *Along the local transition path toward the steady state,* $\dot{\chi}_t^{DD} > 0$ *and* $\dot{H}_t^{DD} > 0$*: the consumption share of GDP rises and the government continues to sell land.*

Proposition 10. *Suppose the long-run land supply is below one and that the following regularity conditions hold for some* τ*: i)* $\dot{Z}_t^{DD} > 0$, $\left(\frac{\partial \mathcal{P}^{DD}}{\partial Z}\right)_t < 0$, $\left(\frac{\partial \mathcal{P}^{DD}}{\partial H}\right)_t < 0$ *for all* $t \geq \tau$*, ii)* $L_t^{DD} \in \left(\frac{\sigma}{\sigma-1}H_s^{DD}, 1\right)$ *for all* $t \geq s \geq \tau$*, and iii)* χ_t^{DD} *is strictly monotonic for all* $t \geq \tau$*. Then, for* $t \geq \tau$*, the consumption share* χ_t^{DD} *strictly rises, the budget-constraint multiplier* λ_t^{DD} *strictly declines, and private land holding and total land supply* H_t^{DD}, L_t^{DD} *both strictly increase.*

Figure 9: Economic growth under second best (SB) and double deviation (DD)



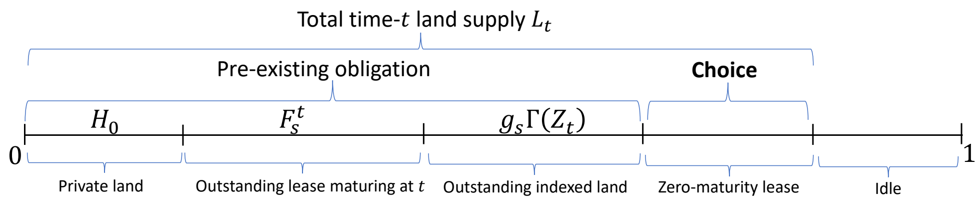
6 Contingent Design of Land Contract

Last, I show that there exists a better design of land contracts that can help restore the SB allocation, under discretion and incomplete markets. Before deriving the result, I first discuss the time inconsistency issue of the no-borrowing implementation of SB. In order to maintain a balanced budget, the government needs to gradually sell land, i.e., following a specific increasing path H_t^{NB}. Suppose the time-s planner wants to reoptimize, if she has the same λ^* as the time-0 planner, she would supply a total amount of land equal to $[1+\eta(\lambda^*)]\, H_s^{NB}$, which will be larger than $L^* = [1+\eta(\lambda^*)]\, H_0$. The reason she does not do this under the no-borrowing implementation is that she keeps a promise made by past self; otherwise the past land price would have been lower. In order for her to still choose L^* when reoptimizing, she needs a specific $\lambda^s > \lambda^*$ to ensure $[1+\eta(\lambda^s)]\, H_s^{NB} = L^*$. However, if she has such a λ^s that is larger than λ^*, that means she is even poorer relative to FB. In that case, she prefers a lower investment rate $\dot{Z}$ and private return than SB.

This reasoning above clearly illustrates the dilemma between balanced budget and time consistency, and indicates that the specific direction of deviation future governments want is to supply less public capital. When markets are complete, the time-0 governments can leave a specific debt portfolio to future governments which will revaluate adversarially if future governments deviate. Can we achieve the same goal via a land contract, rather than requiring the full maturity structure of debt? The answer is yes, and the optimal land contract penalizes deviation by obliging future governments to give more land to household if the stock of public capital Z_t is low, per unit of land contract outstanding.

We could view land contracts as financial securities that pay land (instead of dollar) to the household by the government. A lease (that is valid for an infinitesimal amount of time at present) is a zero-maturity bond. A sale amounts to a consol that pays one unit of land indefinitely. By no means these are the only possibilities. The government can also sell a future lease F^t, which is a right to use one unit of land for an infinitesimal amount of time at future time t. A future lease F^t is like a zero-coupon bond maturing at time t. A full set of future leases spans all bond-like contracts, including current lease and sale. I denote the outstanding amount at time s of a future lease paying off at time t as F_s^t. Further, the government can sell an indexed land contract that pays a variable amount $\Gamma(Z_t)$ at all t depending on the prevailing level of public capital Z_t. I denote its outstanding amount at time s as g_s. Figure 10 illustrates the land allocation, faced by a time-s planner who chooses time-t allocation for all $t \geq s$. The time-s planner inherits pre-existing obligations including private land H_0, outstanding lease maturing at time t of amount F_s^t, and she is obliged to pay a certain amount of land per unit of outstanding indexed land contract, depending on the time-t stock of public capital she chooses.

Figure 10: Land utilization under multiple land contracts



Proposition 11 suggests that an indexed land contract with a specific dependence $\Gamma(\cdot)$ can be combined with future leases to provide incentives for future governments to continue carrying out the time-0 optimal plan.

Proposition 11. (Optimal design of land contract)

1. *When $\frac{1}{\chi_\infty^*} - 1 \geq \frac{\nu(L^*)^{1-\sigma}}{\sigma}$ (or equivalently, $\frac{\sigma}{\sigma-1}H_\infty^{NB} \leq L^*$), the SB allocation can be implemented under double deviation using future leases and Z-indexed land if and only if*

$$\Gamma'(Z_t) = -k\frac{\left(C_t^*\right)^{-1}}{\nu(L^*)^{-\sigma}}\Delta r_t^*$$

with k being an arbitrary positive constant and $\Delta r_t^* \equiv \left[Y'\left(Z_t^*\right) - (\delta+\rho)\right] - \frac{\dot{C}_t^*}{C_t^*} > 0$ capturing the wedge between private return and net marginal product of capital at any given time t along the SB convergence path. The amount outstanding at time t of such Z-indexed land g_s is increasing over time

$$g_s = k^{-1}\left(\frac{1}{\lambda^*} - \frac{1}{\lambda^{s*}}\right)$$
$$F_s^t = \frac{L^*}{1+\eta(\lambda^{s*})} - H_0 - g_s\Gamma(Z_t)$$

2. *When $\frac{1}{\chi_\infty^*} - 1 < \frac{\nu(L^*)^{1-\sigma}}{\sigma}$, the same land contract can achieve SB if the government can borrow using at least one bond.*

The two-part obligation consists of an indexed part $g_s\Gamma(Z_t)$ that penalizes future governments supplying less capital (by requiring them to pay out more land since $\Gamma' < 0$), and future leases F_s^t that make up to H_s^{NB} which is the amount of land sale needed to implement SB with no borrowing. This design essentially splits the rights associated with land sale into two parts, so that it can provide incentives without changing the land income. When

$\frac{\sigma}{\sigma-1}H_{\infty}^{NB} \leq L^*$ is violated, it suffices to have one bond for the government to borrow. This is in contrast to the time-consistent implementation of SB allocation discussed in Section 4, which requires a full maturity structure of debt to provide incentives. Here, the incentive is provided by the indexed land contract.

Last, I provide some remarks on the feasibility of such an indexed land contract. The required legal protection of future leases and indexed land contract is the same as outright sale, in that they are all subject to expropriation. In order for this contract design to work, the indexed land contract has to be honored by future governments. This is a fair requirement, as we note that in order to implement the SB allocation under complete markets à la Lucas and Stokey (1983) and Debortoli, Nunes and Yared (2021), debt contracts issued by the current government have to be acknowledged by future governments too. Further, we note that, while a land contract indexed on capital stock Z_t can help resume the SB allocation, there exist other indexed contracts based on other variables that could achieve the same goal. The advantage of an indexed land contract based on capital stock is that Z_t is easily verifiable, since public investment is an important item in national accounts, and is directly controlled by the government. The timeless dependence $\Gamma(\cdot)$ is easy to enforce too: since it is differentiable, there is no need to argue about a small amount of measurement error in court.

7 Conclusion

Better understanding land finance can help improve policies concerning many developing economies. This paper makes three contributions regarding the mechanism and design of land finance. First, I provide a model to understand land finance and show that the optimal policy should front-load public investment while supplying a constant amount of land over time. Second, I show that the combination of discretion and financial constraints implies that the government will continuously expand land supply, as observed in reality. Lastly, I propose a realistic design of land contracts that links land supply to the provision of public goods to restore the optimal allocation.

This paper focuses on the efficiency aspect of land finance in a deterministic environment. It presents a neoclassical growth model augmented with the role of land finance, which can serve as a starting point for other models. Future research may fruitfully explore the pricing and supply of land in the presence of risk and their role in cyclical fluctuations. If the government is subject to collateral constraints, the use of land finance may create a financial accelerator (Bernanke, Gertler and Gilchrist, 1999) that goes through the public sector rather than the private sector. While this paper studies efficiency in a representative agent model, housing and land play crucial roles in driving wealth inequality across countries (Bonnet et al., 2021). In light of that, how should the policymaker design land finance policies to take into account its distributional impact? Last but not least, this paper addresses a monopoly problem. In practice, land is supplied by many local governments instead of a consolidated government. It is of interest to explore how imperfect competition and migration matter for land finance.

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Online Appendix for

“A Theory of Land Finance and Investment-Led Growth”

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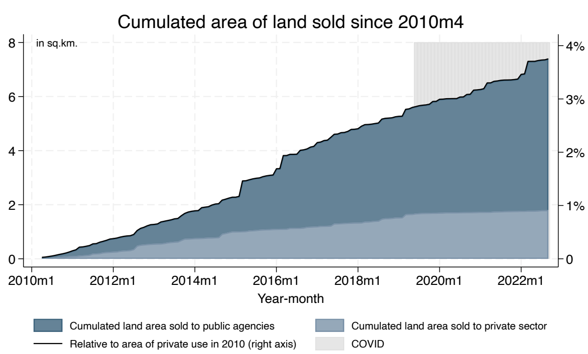
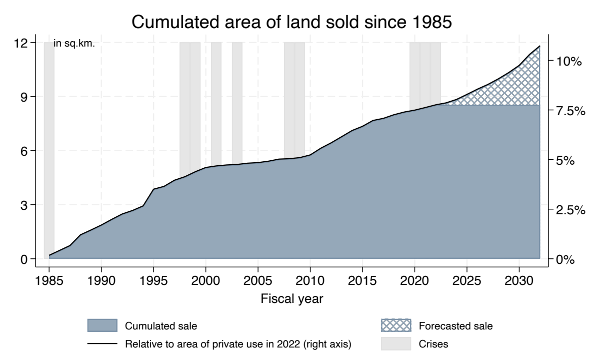
A Additional Figures

Figure A1: Added land supply since beginning of sample

(a) Singapore, 2010-2022



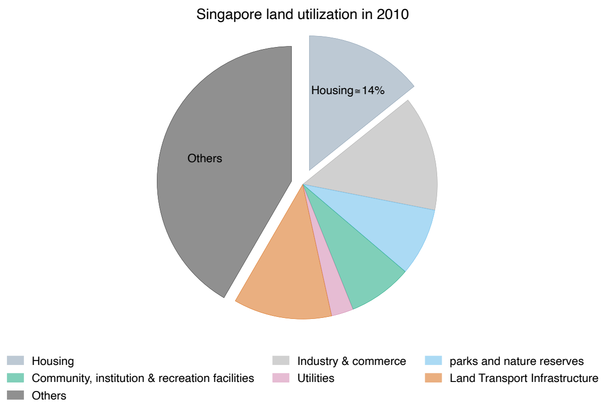
(b) Hong Kong, 1985-2032 (with 10-year forecast)



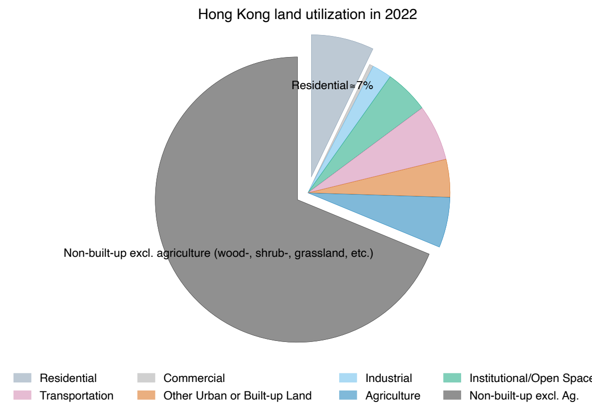
Notes: (a) Singapore data since 2010 are from the aggregate statistics published on Singapore Land Authority’s website. Detailed transfer data by sites going back to last century can be found in websites of various government agencies (Urban Redevelopment Authority, Jurong Town Corporation, and Housing Development Board).
(b) Hong Kong land sale records and forecasts are from the Hong Kong Lands Department.

Figure A2: Land utilization in Singapore, Hong Kong and mainland China

(a) Singapore in 2010



(b) Hong Kong in 2022



(c) Mainland China in 2022

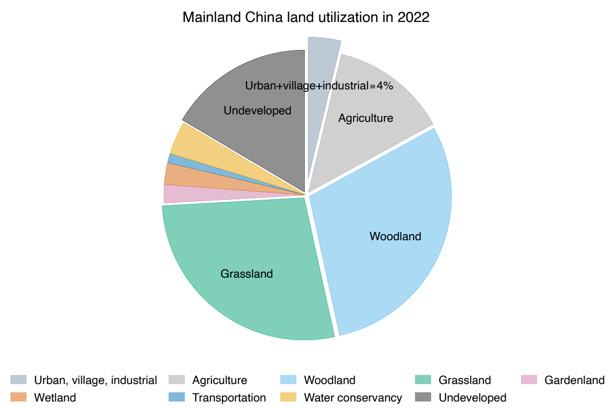
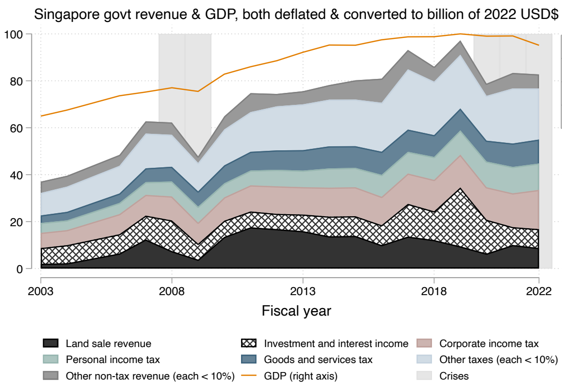
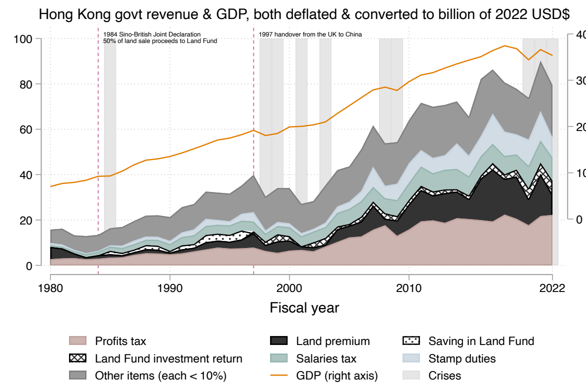


Figure A3: Land sale in government revenue of Asian growth miracles

(a) Singapore, 2003-2022



(b) Hong Kong, 1980-2022



(c) Mainland China, 2001-2022

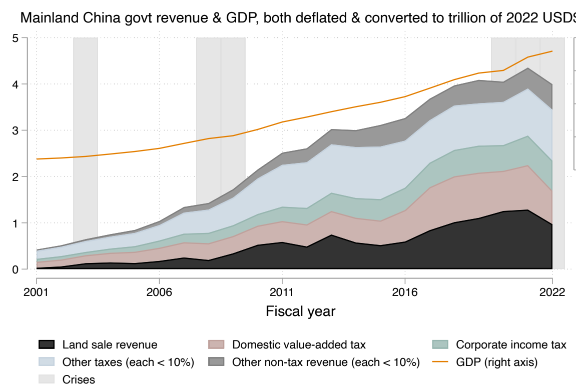
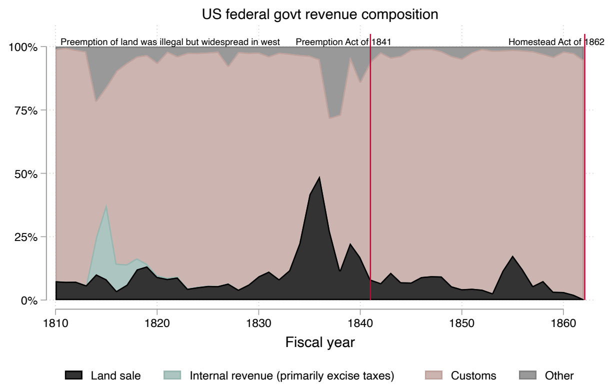
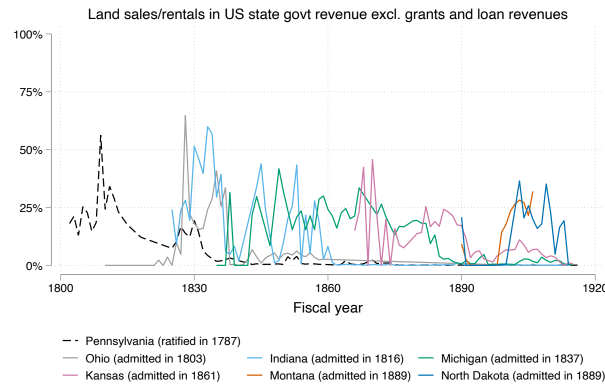


Figure A4: Land sale in US and Hong Kong government revenue in 19th century

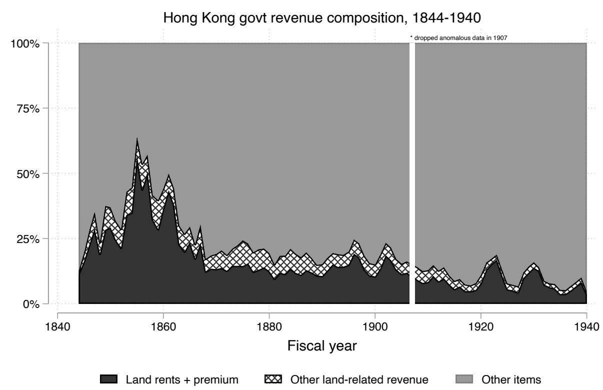
(a) US federal government



(b) US state governments



(c) Hong Kong, 1844-1940



Notes: Data of US federal and state government revenues are from Wallis (2006) and Sylla, Legler and Wallis (1993) respectively.

B Additional Theory Results

B.1 Land Tax and Labor Income Tax

It is known since George (1879) that a tax on land is non-distortionary when land is in fixed supply. I show that this argument may have its limitations as the amount of revenue it raises depends on the total market value of land. Under certain parameters, the government may not raise enough money to support the first-best allocation. In that case, land tax interacts with a motive that restricts supply to increase land value, in order to raise public fundss.

Proposition B1. *Assume that the government can impose a land tax of τ_p. The first best allocation cannot be achieved if*

$$\nu\left(1-\frac{\rho}{\tau_p+\rho}H_0\right)<\int_0^\infty \mathrm{e}^{-\rho t}\left(\frac{1}{\chi_t^{FB}}-1\right)\mathrm{d}t \tag{B1}$$

B.2 Price vs. Quantity Lerner Rule

One may wonder why the requirement of demand elasticity less than one behind Lemma B1 seems to go against the usual intuition that the demand elasticity has to be larger than one to admit a monopoly pricing solution. Let us indeed consider such a textbook example (Tirole, 1988, Chapter 1.1): a monopoly quotes a price p under constant marginal cost mc and demand function $q(p)$ to maximize profits,

$$\max \pi = (p-mc)\underbrace{q(p)}_{\text{demand}} \tag{B2}$$

The optimal price satisfies the familiar Lerner condition

$$\frac{p-mc}{p}=\left(-\frac{\partial \log q(p)}{\partial \log p}\right)^{-1} \tag{B3}$$

The demand elasticity $-\frac{\partial \log q}{\partial \log p}$ has to be larger than one to admit an interior solution. We observe that (B3) and (B5) are isomorphic once we swap the price and quantity. In the problem (B4), the price declines in supply, but the monopoly does not profit from the pre-owned amount q_0, which acts to erode the profit in a way much like the marginal cost in problem (B2). Thus the requirement for (B4) to admit a solution is exactly the opposite: the demand for housing has to be inelastic rather than elastic, which is also empirically relevant. Note that the key difference is not in whether the monopoly sets price or quantity, which maps into each other via the demand function $q(p)$, but rather where a quantity-like term (q_0) or a price-like term (mc) erodes profits.

Assumption 2 ensures that the FB allocation cannot be achieved with maximum land supply $L_t = 1$. I further focus on a case that is empirically relevant where the government can raise more money by restricting land supply.

I first take a small detour to derive a useful variant of the celebrated Lerner rule, which will also provide useful intuition for our later results.

Lemma B1. (A quantity Lerner rule) *Consider a monopoly that faces a demand curve $q(p)$. Assume that it produces with zero marginal cost, but the buyer has already owned some quantity q_0, so the monopoly profit is*

$$\max \pi = (q(p) - q_0)\, p \tag{B4}$$

The supply to maximize profit π satisfies

$$\frac{q - q_0}{q} = \underbrace{-\frac{\mathrm{d}\log q(p)}{\mathrm{d}\log p}}_{\text{demand elasticity}} \tag{B5}$$

Formula (B5) suggests that, in problem (B4), a solution with finite quantity exists when the demand is *inelastic*. This is meaningful as decades of empirical research almost all point to inelastic demand for housing.[10] This requirement is in contrast to the canonical Lerner rule that gives rise to a solution only when demand is elastic and characterizes price relative to marginal cost. The high-level difference is that it is a pre-owned quantity q_0 that erodes the monopoly's profit in problem (B4), rather than a positive marginal cost as in the textbook case. Appendix B.2 discusses this in greater detail. While our quantity Lerner rule works under inelastic demand and the price Lerner rule is only applicable to elastic demand, they share qualitatively similar comparative statics: as the demand elasticity rises, the supply q increases and the price p decreases.

Coming back to our analysis of land finance, taking household demand of housing (8) and holding fixed non-durable consumption C_t, we can derive the demand elasticity of housing

$$-\frac{\partial \log L_t}{\partial \log D_t} = \sigma^{-1}$$

which is less than one under our assumption $\sigma > 1$. That is, an ϵ percentage demand shock to the housing market changes the price of housing by $\sigma\epsilon$ percentage. If the government were to maximize its time-t profit from land lease, i.e., $\max_{L_t} (L_t - H_t)\, D_t(L_t)$, the optimal land supply would follow from Lemma B1, if interior,

$$\frac{L_{mono} - H_t}{L_{mono}} = -\frac{\partial \log L_t}{\partial \log D_t} = \sigma^{-1} \quad \Rightarrow \quad L_{mono} = \frac{\sigma}{\sigma - 1} H_t \tag{B6}$$

Thus a government that maximizes profits would set $L_{mono} = \frac{\sigma}{\sigma-1} H_t$ if $\frac{\sigma}{\sigma-1} H_t < 1$. This

[10] Albouy, Ehrlich and Liu (2016) estimate an elasticity of about two thirds and review earlier estimates.

suggests that the total supply will be proportional to the outstanding amount H_t under constant elasticity, which is parallel to the price Lerner rule that prescribes a price proportional to the marginal cost.

Later, when I introduce a benevolent government that maximizes the household's utility, it would balance the need to raise money for investment against the household utility gain from living in larger houses. Relative to this profit-maximizing benchmark, it will supply more land to the household.

Henceforth I assume that $\frac{\sigma}{\sigma-1}H_0 < 1$, which implies the government can profit from restricting land supply.[11]

B.3 Time-Consistent Implementation with Government as Net Creditor

Recalling that any no-borrowing implementation of the SB allocation characterized in Proposition 6 has the property that the government gradually sells land, we establish the following impossibility result.

Proposition B2. *There exists an implementation of the SB allocation that is both time-consistent and involving non-positive* total value *of government debt (i.e. government as net creditor), if and only if*

$$\frac{1}{\chi_\infty^*} - 1 \geq \frac{\nu\,(L^*)^{1-\sigma}}{\sigma}. \tag{B7}$$

If δ is small (i.e., $\frac{\alpha\delta}{(1-\alpha)\delta+\rho} \leq \frac{\nu}{\sigma}$), this inequality never holds. Otherwise it holds only when λ^* is small.

Intuitively, if infrastructure never depreciates ($\delta = 0$), in any time-consistent implementation, the government needs no fiscal income at the steady state. Thus it should put all land supply in private hands ($H_\infty^{NB} = L^*$), since it has no debt to repay either. In this case, the SB allocation can never be implemented in a time-consistent way, since $H_\infty^{NB} > \frac{\sigma-1}{\sigma}L^*$.

If δ is large, the government needs lease income at the steady state to replenish infrastructure. When λ^* is large, the steady-state consumption share χ_∞^* is high and investment ratio is low, suggesting a low need of fiscal income. Further, as λ^* is large, the land supply is depressed and the land income is high, the government needs to hold less land to derive a certain amount of income. Taking together, the government chooses H_∞ that exceeds $\frac{\sigma-1}{\sigma}L^*$ when λ^* is large enough, denying (B7).

[11] This assumption and (11) can hold simultaneously as they involve different parameters.

I conclude this section with a remark that, in cases where (B7) does hold, even though the total value of government debt is non-positive, the government needs the whole maturity structure to implement SB allocation in a time-consistent way, based on Proposition 7.

C Derivations and Proofs

C.1 Optimality and Implementability Conditions

Derivation of household optimality conditions (7-9).

$$\begin{aligned}
\max & \int_0^\infty \mathrm{e}^{-\rho t} U\left(C_t\right) \mathrm{d}t \\
\text{s. t.} & \int_0^\infty Q_t \left[C_t - Y_t + P_t \dot{H}_t + D_t \left(L_t - H_t\right)\right] \mathrm{d}t = 0
\end{aligned}$$

where $Q_0 = 1$. Integrate the budget constraint by parts to get

$$\begin{aligned}
0 &= \int_0^\infty Q_t \left[C_t - Y_t + P_t \dot{H}_t + D_t \left(L_t - H_t\right)\right] \mathrm{d}t \\
&= \int_0^\infty Q_t \left[C_t - Y_t + D_t \left(L_t - H_t\right)\right] \mathrm{d}t - P_0 H_0 - \int_0^\infty H_t \mathrm{d}\left(P_t Q_t\right)
\end{aligned}$$

Thus the FOCs w.r.t. C_t, L_t, H_t are

$$\begin{aligned}
\frac{\mathrm{e}^{-\rho t} U_{C,t}}{U_{C,0}} &= Q_t \\
\frac{U_{L,t}}{U_{C,t}} &= D_t \\
\mathrm{d}\left(P_t Q_t\right) &= -Q_t D_t \mathrm{d}t
\end{aligned}$$

We iterate the last expression to get

$$\begin{aligned}
P_t &= Q_t^{-1} \int_t^\infty Q_s D_s \mathrm{d}s \\
&= \frac{U_{C,0}}{\mathrm{e}^{-\rho t} U_{C,t}} \int_t^\infty \frac{\mathrm{e}^{-\rho s} U_{C,s}}{U_{C,0}} \frac{U_{L,s}}{U_{C,s}} \mathrm{d}s \\
&= U_{C,t}^{-1} \mathcal{P}_t
\end{aligned}$$

with $\mathcal{P}_t \equiv \int_t^\infty \mathrm{e}^{-\rho(s-t)} U_{L,s} \mathrm{d}s$.□

Proof of Lemma B1. The first-order condition is

$$0 = \frac{\mathrm{d}\pi}{\mathrm{d}q} = p + (q - q_0) \frac{\mathrm{d}p}{\mathrm{d}q}$$

and thus

$$\frac{q}{q - q_0} = -\frac{\mathrm{d}\log p}{\mathrm{d}\log q}$$

□

Derivation of implementability constraint (12). Plugging (7, 8, 9) into (6) and setting $T_t = 0$ yields

$$\begin{aligned}
0 &= \int_0^\infty \mathrm{e}^{-\rho t} U_{C,0}^{-1} \left[(C_t - Y_t) U_{C,t} + \dot{H}_t \mathcal{P}_t + (L_t - H_t) U_{L,t} \right] \mathrm{d}t \\
0 &= \int_0^\infty \mathrm{e}^{-\rho t} \left[(C_t - Y_t) U_{C,t} + (L_t - H_t) U_{L,t} \right] \mathrm{d}t - H_0 \boldsymbol{\mathcal{U}}_{L,0} - \int_0^\infty H_t \mathrm{d}\left(\mathrm{e}^{-\rho t} \mathcal{P}_t \right) \\
&= \int_0^\infty \mathrm{e}^{-\rho t} \left[(C_t - Y_t) U_{C,t} + L_t U_{L,t} \right] \mathrm{d}t - H_0 \boldsymbol{\mathcal{U}}_{L,0} \\
&= \int_0^\infty \mathrm{e}^{-\rho t} \left[(C_t - Y_t) U_{C,t} + (L_t - H_0) U_{L,t} \right] \mathrm{d}t
\end{aligned}$$

□

Present-value sum of public capital, useful for (13). I establish a relation that will be helpful in forming a Lagrangian instead of a Hamiltonian, to make the continuous-time model more intuitive and relatable to a discrete-time model. Take the continuous-time resource constraint (2)

$$0 = Y_t(Z_t) - C_t - \delta Z_t - \dot{Z}_t,$$

attach an arbitrary multiplier $\{\mu_t \mathrm{e}^{-\rho t}\}_{t \geq 0}$, and integrate by parts to get

$$\begin{aligned}
0 &= \int_0^\infty \mu_t \mathrm{e}^{-\rho t} \left[Y_t(Z_t) - C_t - \delta Z_t - \dot{Z}_t \right] \mathrm{d}t \\
&= \int_0^\infty \mu_t \mathrm{e}^{-\rho t} \left[Y_t(Z_t) - C_t - \delta Z_t \right] \mathrm{d}t - \left[\mu_t \mathrm{e}^{-\rho t} Z_t |_0^\infty - \int Z_t \mathrm{d}\left(\mu_t \mathrm{e}^{-\rho t} \right) \right]
\end{aligned}$$

$$= \int_0^\infty \mathrm{e}^{-\rho t} \left\{ \mu_t \left[Y_t \left(Z_t \right) - C_t - \left(\delta + \rho \right) Z_t \right] + \dot{\mu}_t Z_t \right\} \mathrm{d}t + \mu_0 Z_0. \tag{C1}$$

This is a present-value constraint that depends on Z_t but not $\dot{Z}_t$.□

C.2 Proofs

Proof of Proposition 1.

$$\begin{aligned} \max \mathcal{L}^{FB} &= \int_0^\infty \mathrm{e}^{-\rho t} U \left(C_t, L_t \right) \mathrm{d}t + \int_0^\infty \mu_t \mathrm{e}^{-\rho t} \left[Y_t \left(Z_t \right) - C_t - \delta Z_t - \dot{Z}_t \right] \mathrm{d}t \\ &= \int_0^\infty \mathrm{e}^{-\rho t} U \left(C_t, L_t \right) \mathrm{d}t + \int_0^\infty \mathrm{e}^{-\rho t} \left\{ \mu_t \left[Y_t \left(Z_t \right) - C_t - \left(\delta + \rho \right) Z_t \right] + \dot{\mu}_t Z_t \right\} \mathrm{d}t + \mu_0 Z_0 \end{aligned}$$

The optimality conditions are

$$\begin{aligned} 0 &= \mathrm{e}^{\rho t} \frac{\partial \mathcal{L}^{FB}}{\partial C_t} = U_{C,t} - \mu_t \\ 0 &= \mathrm{e}^{\rho t} \frac{\partial \mathcal{L}^{FB}}{\partial Z_t} = \dot{\mu}_t + \left[Y_t' - \left(\delta + \rho \right) \right] \mu_t \end{aligned}$$

Thus, the consumption growth rate is

$$\frac{\dot{C}_t}{C_t} = Y_t' - \left(\delta + \rho \right)$$

Because U_L is strictly positive, the upper land bound binds at every date, so $L_t^{FB} = 1$. Because H_t does not appear in the planner objective or the remaining constraints, any feasible $H_t^{FB} \in [0, 1]$ is optimal.□

Proof of Proposition 2. Since the government can give lump-sum rebates but cannot collect lump-sum tax, the first best cannot be implemented if the present value of the land income is lower than the present value of the FB investment, i.e., $INC^{FB} < INV^{FB}$, which amounts to (10).

Next, the left-hand side is strictly decreasing in H_0, so the condition is more likely to hold when the government starts with less land to monetize.

For Z_0, fix all parameters other than the initial infrastructure stock and consider two initial conditions $Z_0' < Z_0$. Along the convergent first-best saddle path, the economy with lower Z_0 starts earlier on the same path, so

$$\chi_t^{FB}(Z_0') < \chi_t^{FB}(Z_0) \qquad \text{for all } t \geq 0.$$

which implies that the right-hand side is larger when Z_0 is lower. Thus the condition is also more likely to hold when initial infrastructure is low.

The LHS of (10) obtains its maximum of ν with $H_0 = 0$. Its RHS is weakly larger than $\rho \int_0^\infty \mathrm{e}^{-\rho t} \left(\frac{1}{\chi_\infty^{FB}} - 1 \right) \mathrm{d}t = \frac{1}{\chi_\infty^{FB}} - 1 = \frac{\delta \alpha}{\delta(1-\alpha)+\rho}$, since $\chi_t^{FB} \leq \chi_\infty^{FB}$ with the equality obtained when the economy starts with $Z_0 = Z_\infty^{FB}$ under first best. Hence (11) is a sufficient condition for (10).□

Proof of Proposition 3. Because Proposition 2 rules out first-best implementability, the Ramsey multiplier must satisfy$\lambda^* > 0$.

The FOCs are

$$0 = \mathrm{e}^{\rho t} \frac{\partial \mathcal{L}^*}{\partial C_t} = (1 + \lambda^*) U_{C,t} + \lambda^* (C_t - Y_t) U_{CC,t} - \mu_t^* = U_{C,t} - \lambda^* Y_t U_{CC,t} - \mu_t^* \tag{C2}$$

$$0 = \mathrm{e}^{\rho t} \frac{\partial \mathcal{L}^*}{\partial L_t} = (1 + \lambda^*) U_{L,t} + \lambda^* (L_t - H_0) U_{LL,t} - \zeta_t^* \tag{C3}$$

$$0 = \mathrm{e}^{\rho t} \frac{\partial \mathcal{L}^*}{\partial Z_t} = \mu_t^* [Y_t' - (\delta + \rho)] + \dot{\mu}_t^* - \lambda^* Y_t' U_{C,t} \tag{C4}$$

Consumption growth and fiscal balance. Use (C2) to get

$$\mu_t^* = (1 + \lambda^* Y_t / C_t) C_t^{-1} = \left(1 + \frac{\lambda^*}{\chi_t}\right) C_t^{-1}$$

Plug this into (C4) and arrive at

$$0 = \left(1 + \frac{\lambda^*}{\chi_t}\right) C_t^{-1} [Y_t' - (\delta + \rho)] - \left(1 + \frac{\lambda^*}{\chi_t}\right) C_t^{-2} \dot{C}_t - \frac{\lambda^*}{(\chi_t)^2} C_t^{-1} \dot{\chi}_t - \lambda^* Y_t' C_t^{-1}$$

which simplifies to

$$\frac{\dot{C}_t^*}{C_t^*} = [Y_t'(Z_t^*) - (\delta + \rho)] - \frac{\lambda^*}{\lambda^* + \chi_t^*} \left[\chi_t^* Y_t'(Z_t^*) + \frac{\dot{\chi}_t^*}{\chi_t^*} \right]$$

For any (Z_0, H_0), the multiplier λ^* satisfies the implementability condition (12). I write

(12) as equalizing fiscal income and expenditure in unit of utility

$$\underbrace{\int_0^\infty \mathrm{e}^{-\rho t}\left(L_t - H_0\right) U_{L,t}\mathrm{d}t}_{\equiv INC^*} = \underbrace{\int_0^\infty \mathrm{e}^{-\rho t}\left(Y_t - C_t\right) U_{C,t}\mathrm{d}t}_{\equiv INV^*} \tag{C5}$$

When $\lambda^* > \frac{1}{\sigma(1-H_0)-1}$, the optimal land supply is interior. In this case, the present-value util-unit sum of fiscal income INC^* equals

$$INC^* = \int_0^\infty \mathrm{e}^{-\rho t}\eta H_0 U_L\left(\left(1+\eta\right)H_0\right)\mathrm{d}t = \frac{\nu H_0^{1-\sigma}}{\rho}\frac{\eta}{\left(1+\eta\right)^\sigma}$$

which only depends on (H_0, λ^*), i.e., $INC^* = INC^*\left(H_0, \lambda^*\right)$. It satisfies $\frac{\partial INC^*}{\partial H_0} < 0$ and

$$\frac{\partial INC^*}{\partial \lambda^*} = \frac{\nu H_0^{1-\sigma}}{\rho}\frac{1-(\sigma-1)\eta}{\left(1+\eta\right)^{\sigma+1}}\frac{\mathrm{d}\eta}{\mathrm{d}\lambda^*} = -\frac{\nu H_0^{1-\sigma}}{\rho}\left(1+\eta\right)^{-(\sigma+1)}\frac{\eta\sigma}{1+\lambda^*}\frac{\mathrm{d}\eta}{\mathrm{d}\lambda^*} > 0$$

Intuitively, the larger the need for fiscal income (higher λ^*), the larger the fiscal income (INC^*). INC^* is independent of Z_0 except through its dependence on λ^*.

The present-value util-unit sum of investment INV^* equals

$$INV^* = \int_0^\infty \mathrm{e}^{-\rho t}\left(\frac{1}{\chi_t} - 1\right)\mathrm{d}t$$

which implicitly depends on Z_0 and λ^* through the ODE system (2, 15), i.e., $INV^* = INV^*\left(Z_0, \lambda^*\right)$.

Phase diagram in (Z, C) **space.** The $\dot{Z} = 0$ locus is from (2) as

$$C_t^* = Y\left(Z_t^*\right) - \delta Z_t^*$$

which is concave and positive between 0 and $\left(\frac{A}{\delta}\right)^{\frac{1}{1-\alpha}}$, the same as in the neoclassical growth model. Its apex value is $C^a = \left(\alpha^{\frac{\alpha}{1-\alpha}} - \alpha^{\frac{1}{1-\alpha}}\right)\left(\frac{A}{\delta^\alpha}\right)^{\frac{1}{1-\alpha}}$ at $Z^a = \left(\frac{\alpha A}{\delta}\right)^{\frac{1}{1-\alpha}}$.

To arrive at the $\dot{C} = 0$ locus, note that

$$\frac{\dot{\chi}_t^*}{\chi_t^*} = \frac{\dot{C}_t^*}{C_t^*} - \frac{\dot{Y}_t^*}{Y_t^*} = \frac{\dot{C}_t^*}{C_t^*} - \alpha\frac{\dot{Z}_t^*}{Z_t^*} = \frac{\dot{C}_t^*}{C_t^*} - \alpha\frac{Y_t^* - C_t^* - \delta Z_t^*}{Z_t^*}$$

where I have plugged in the production function and (2). Then, using (15) and setting $\dot{C}_t^* = 0$, I get

$$\begin{aligned} 0 &= Y'\left(Z_t^*\right) - (\delta + \rho) - \frac{\lambda^*}{\lambda^* + C_t^*/Y\left(Z_t^*\right)}\left[\frac{C_t^*}{Y\left(Z_t^*\right)} Y'\left(Z_t^*\right) - \alpha \frac{Y\left(Z_t^*\right) - C_t^* - \delta Z_t^*}{Z_t^*}\right] \\ &= Y'\left(Z_t^*\right) - (\delta + \rho) + \frac{\alpha\lambda^*}{\lambda^* + C_t^*/Y\left(Z_t^*\right)} \frac{Y\left(Z_t^*\right) - 2C_t^* - \delta Z_t^*}{Z_t^*} \end{aligned} \tag{C6}$$

In the case with $\lambda^* = 0$, this locus is $0 = Y'\left(Z_t^*\right) - (\delta + \rho)$ which specifies a constant $Z_t^* \equiv Z_\infty^{FB} = \left(\frac{\alpha A}{\delta+\rho}\right)^{\frac{1}{1-\alpha}}$ (a vertical line in (Z, C) space) as in the first best. Its intersection with the $\dot{Z} = 0$ locus is $C_\infty^{FB} = Y\left(Z_\infty^{FB}\right) - \delta Z_\infty^{FB}$. For any $\lambda^* > 0$, the $\dot{C} = 0$ locus is no longer a vertical line at Z_∞^{FB}, but it will cross that vertical line once at exactly $(Z_\infty^{FB}, C_\infty^{FB}/2)$.[12] I will prove in Proposition 4 that the steady state C_∞^*, Z_∞^* both decrease in λ^*, suggesting that the intersection of $\dot{Z} = 0$ locus and $\dot{C} = 0$ locus under $\lambda^* > 0$ is to the left of Z_∞^{FB}. That is, the $\dot{C} = 0$ locus is rotated counterclockwise around $(Z_\infty^{FB}, C_\infty^{FB}/2)$ for $\lambda^* > 0$.

Monotonicity of consumption share χ_t. Omit the time subscript and $*$ superscript here for simplicity. We have

$$\begin{aligned} g^\chi \equiv \frac{\dot{\chi}}{\chi} &= \frac{\dot{C}}{C} - \frac{\dot{Y}}{Y} = \frac{\dot{C}}{C} - \alpha\frac{\dot{Z}}{Z} = \frac{\dot{C}}{C} - \alpha\frac{Y - C - \delta Z}{Z} \\ &= \left[Y' - (\delta + \rho)\right] - \frac{\lambda}{1 + \frac{\lambda}{\chi}}\left[\frac{\dot{\chi}}{\chi^2} + Y'\right] - Y'(1 - \chi) + \alpha\delta \\ &= Y'\chi - \left[(1 - \alpha)\delta + \rho\right] - \frac{\lambda}{1 + \frac{\lambda}{\chi}}\left(\frac{\dot{\chi}}{\chi^2} + Y'\right) \\ &= Y'\frac{\chi^2}{\chi + \lambda} - \left[(1 - \alpha)\delta + \rho\right] - \frac{\lambda}{\chi + \lambda} g^\chi \\ \frac{\chi + 2\lambda}{\chi + \lambda} g^\chi &= Y'\frac{\chi^2}{\chi + \lambda} - \left[(1 - \alpha)\delta + \rho\right] \end{aligned} \tag{C7}$$

[12] To be precise, there is only one crossing in the $\dot{Z} > 0$ subspace ($C \leq Y(Z) - \delta Z$). Outside that, there is another uninteresting crossing at $C \to \infty$ which corresponds to infinitely fast depreciation of Z.

Let $\gamma \equiv \frac{\chi+2\lambda}{\chi+\lambda} g^{\chi}$ which has the same sign as g^{χ}. We have

$$\begin{aligned}
\gamma &= Y' \frac{\chi^2}{\chi+\lambda} - [(1-\alpha)\delta + \rho] \\
\dot{\gamma} &= Y'' \dot{Z} \frac{\chi^2}{\chi+\lambda} + Y' \frac{(\chi+2\lambda)\chi^2}{(\chi+\lambda)^2} g^{\chi} \\
&= Y'' \dot{Z} \frac{\chi^2}{\chi+\lambda} + Y' \frac{\chi^2}{\chi+\lambda} \gamma
\end{aligned}$$

If $\gamma \leq 0$ for some t, then $\dot{\gamma} < 0$ since $Y'' < 0, \dot{Z} > 0, Y' > 0$ at that time, as Z monotonically increases when starting from the left of Z_∞ in the third quadrant on the phase diagram. It suggests that $\gamma < 0$ holds forever and γ keeps decreasing, a result inconsistent with convergence. Thus we must have $\gamma > 0$ for all t, which means $g^{\chi} > 0$ and thus $\dot{\chi} > 0$.□

Proof of Proposition 4.

Steady state and comparative statics w.r.t. λ^***.** At the long-run steady state, (15) gives

$$\left(1 - \frac{\lambda^*}{1 + \frac{\lambda^*}{\chi_\infty^*}}\right) Y'_\infty = \delta + \rho \tag{C8}$$

Case with no depreciation. If $\delta = 0$, then $\chi_\infty^* = 1$ from the resource constraint and thus (C8) reduces to $Y'_\infty = (1+\lambda^*)\rho$. Hence

$$Z_\infty^* = (Y')^{-1}((1+\lambda^*)\rho) = \left[\frac{\alpha}{(1+\lambda^*)\rho}\right]^{\frac{1}{1-\alpha}}$$

and $C_\infty^* = Y_\infty^* = A(Z_\infty^*)^{\alpha}$. Obviously, both decrease in λ^*.

Case with positive depreciation. If $\delta > 0$, the resource constraint implies that $Z_\infty^* = \frac{Y_\infty^* - C_\infty^*}{\delta}$ and thus the marginal product is

$$Y'_\infty = \frac{\alpha Y_\infty^*}{Z_\infty^*} = \frac{\alpha\delta}{1-\chi_\infty^*} \tag{C9}$$

Plugging this in (C8) gives

$$\left(1-\frac{\lambda^*}{1+\frac{\lambda^*}{\chi^*_\infty}}\right)\frac{\alpha\delta}{1-\chi^*_\infty}=\delta+\rho$$
$$\left(1+\frac{\chi^*_\infty}{1-\chi^*_\infty-\lambda^*+\frac{\lambda^*}{\chi^*_\infty}}\right)\alpha\delta=\delta+\rho$$
$$\frac{\chi^*_\infty}{1-\chi^*_\infty-\lambda^*+\frac{\lambda^*}{\chi^*_\infty}}=\frac{\delta\,(1-\alpha)+\rho}{\alpha\delta}$$

It is a quadratic equation in χ^*

$$(\delta+\rho)\,(\chi^*_\infty)^2-(1-\lambda^*)\,[\delta\,(1-\alpha)+\rho]\,\chi^*_\infty-\lambda^*\,[\delta\,(1-\alpha)+\rho]=0$$

which gives two solutions

$$\chi^*_\infty=\frac{(1-\lambda^*)\,[\delta\,(1-\alpha)+\rho]\pm\sqrt{(1-\lambda^*)^2\,[\delta\,(1-\alpha)+\rho]^2+4\,(\delta+\rho)\,\lambda^*\,[\delta\,(1-\alpha)+\rho]}}{2\,(\delta+\rho)}$$
$$=\frac{(1-\lambda^*)\pm\sqrt{(1-\lambda^*)^2+4\frac{\delta+\rho}{\delta\,(1-\alpha)+\rho}\lambda^*}}{2}\frac{\delta\,(1-\alpha)+\rho}{\delta+\rho}$$
$$=\frac{(1-\lambda^*)\pm\sqrt{(1+\lambda^*)^2+4\frac{\delta\alpha}{\delta\,(1-\alpha)+\rho}\lambda^*}}{2}\frac{\delta\,(1-\alpha)+\rho}{\delta+\rho}$$

We take the solution with plus sign since consumption can only be positive.

χ^*_∞ increases in λ^* since

$$\frac{\mathrm{d}\chi^*_\infty}{\mathrm{d}\lambda^*}\propto-1+\frac{(1+\lambda^*)+2\frac{\delta\alpha}{\delta(1-\alpha)+\rho}}{\sqrt{(1+\lambda^*)^2+4\frac{\delta\alpha}{\delta\,(1-\alpha)+\rho}\lambda^*}}>0$$

(C9) gives infrastructure

$$Z^*_\infty=(Y')^{-1}\left(\frac{\alpha\delta}{1-\chi^*_\infty}\right)=\left[\frac{A\,(1-\chi^*_\infty)}{\delta}\right]^{\frac{1}{1-\alpha}}$$

which decreases in χ^*_∞ and thus decreases in λ^*.

Further, the consumption is

$$C^*_\infty = \chi^*_\infty Y^*_\infty = A\chi^*_\infty \left[\frac{A\left(1-\chi^*_\infty\right)}{\delta}\right]^{\frac{\alpha}{1-\alpha}}$$

To show that it is decreasing in λ^*, it suffices to show that $\chi^*_\infty\left(1-\chi^*_\infty\right)^{\frac{\alpha}{1-\alpha}}$ decreases in χ^*_∞. We check

$$\frac{\mathrm{d}\log\left[\chi^*_\infty\left(1-\chi^*_\infty\right)^{\frac{\alpha}{1-\alpha}}\right]}{\mathrm{d}\chi^*_\infty} = \frac{1}{\chi^*_\infty} - \frac{\alpha}{1-\alpha}\frac{1}{1-\chi^*_\infty} = \frac{1-\alpha-\chi^*_\infty}{(1-\alpha)\left(1-\chi^*_\infty\right)\chi^*_\infty}$$

which is negative since $\chi^*_\infty > \frac{\delta(1-\alpha)+\rho}{\delta+\rho} > 1-\alpha$.

Comparative statics w.r.t. initial conditions H_0, Z_0. We start with a relaxed Ramsey problem that nests our problem. Fix initial conditions (Z_0, H_0). Let $\mathcal{F}(Z_0)$ denote the set of feasible paths $\{C_t, L_t, Z_t\}_{t\geq 0}$ satisfying the resource constraint (2), the initial condition $Z(0) = Z_0$, the land constraint $0 \leq L_t \leq 1$, and the transversality condition. Using the implementability condition derived in (C5), define, for each candidate $\lambda \geq 0$, the *relaxed Ramsey problem*

$$\begin{aligned} G(\lambda; Z_0, H_0) \equiv \max_{\{C_t,L_t,Z_t\}\in\mathcal{F}(Z_0)} \Bigg\{ \int_0^\infty & \mathrm{e}^{-\rho t} U(C_t, L_t)\,\mathrm{d}t \\ & + \lambda\left[\int_0^\infty \mathrm{e}^{-\rho t}(L_t - H_0)U_{L,t}\,\mathrm{d}t - \int_0^\infty \mathrm{e}^{-\rho t}(Y_t - C_t)U_{C,t}\,\mathrm{d}t\right]\Bigg\}. \end{aligned} \tag{C10}$$

In words, the original implementability condition is not imposed as a hard equality constraint; instead, its residual is added to the objective with weight λ.

For each fixed λ, let $\{C^\lambda_t, L^\lambda_t, Z^\lambda_t\}_{t\geq 0}$ denote the optimizer of (C10). Define

$$INC(\lambda, H_0) \equiv \int_0^\infty \mathrm{e}^{-\rho t}(L^\lambda_t - H_0)U^\lambda_{L,t}\,\mathrm{d}t,$$

$$INV(Z_0, \lambda) \equiv \int_0^\infty \mathrm{e}^{-\rho t}(Y^\lambda_t - C^\lambda_t)U^\lambda_{C,t}\,\mathrm{d}t.$$

Under log utility, $U^\lambda_{C,t} = 1/C^\lambda_t$ and $\chi_t(Z_0, \lambda) \equiv C^\lambda_t / Y^\lambda_t$, so the second term can also be written as

$$INV(Z_0, \lambda) = \int_0^\infty \mathrm{e}^{-\rho t}\left(\frac{1}{\chi_t(Z_0, \lambda)} - 1\right)\mathrm{d}t.$$

Finally, define the implementability residual

$$R(\lambda; Z_0, H_0) \equiv INC(\lambda, H_0) - INV(Z_0, \lambda).$$

If $R(\lambda; Z_0, H_0) > 0$, then fiscal income exceeds fiscal expenditure in utility units under the relaxed optimum; if $R(\lambda; Z_0, H_0) < 0$, the opposite is true. The true Ramsey multiplier $\lambda^*(Z_0, H_0)$ is exactly the value of λ that restores the implementability condition, that is,

$$R(\lambda^*(Z_0, H_0); Z_0, H_0) = 0.$$

For every fixed feasible allocation, the objective in (C10) is affine in λ. Therefore $G(\lambda; Z_0, H_0)$, being the pointwise maximum of affine functions, is convex in λ. By the envelope theorem,

$$G_\lambda(\lambda; Z_0, H_0) = R(\lambda; Z_0, H_0)$$

wherever the derivative exists. Since the derivative of a convex function is weakly increasing, it follows that $R(\lambda; Z_0, H_0)$ is weakly increasing in λ.

Effect of H_0 on λ^*. Fix Z_0 and consider two initial land holdings $H_0 < H_0'$. For a given λ, the dynamic system in (15) and (2) does not contain H_0. Therefore the path of $\chi_t(Z_0, \lambda)$, and hence $INV(Z_0, \lambda)$, is independent of H_0.

By contrast, H_0 enters the fiscal-income term directly through land supply. And $INC(\lambda, H_0)$ is continuous and strictly decreasing in H_0 for every fixed λ.

Since $INV(Z_0, \lambda)$ does not depend on H_0, we obtain

$$R(\lambda; Z_0, H_0') < R(\lambda; Z_0, H_0) \qquad \text{for all } \lambda \text{ whenever } H_0' > H_0.$$

Now evaluate this inequality at the old root $\lambda^*(Z_0, H_0)$. Because $R(\lambda^*(Z_0, H_0); Z_0, H_0) = 0$, we get

$$R(\lambda^*(Z_0, H_0); Z_0, H_0') < 0.$$

Since R is increasing in λ, the value of λ needed to bring the residual back to zero must be larger. Therefore

$$\lambda^*(Z_0, H_0') > \lambda^*(Z_0, H_0).$$

Thus λ^* is strictly increasing in H_0.

Effect of Z_0 on λ^*. Fix H_0 and consider two initial infrastructure stocks $Z_0' < Z_0$. For a given λ, the fiscal-income term $INC(\lambda, H_0)$ does not depend on Z_0. So it remains to study the expenditure term $INV(Z_0, \lambda)$.

Here the autonomous nature of the dynamic system is crucial. For a fixed λ, the system in (15) and (2) is autonomous, so the convergent solutions all lie on the same saddle path. Along the relevant convergent branch considered in Proposition 3, the consumption share χ_t is strictly increasing over time for any λ. Therefore changing Z_0 does not create a new trajectory; it only changes where the economy starts on the same trajectory. In particular, if $Z_0' < Z_0$, then the economy with lower initial infrastructure starts earlier on that saddle path. Hence there exists some $\tau > 0$ such that

$$\chi_t(Z_0, \lambda) = \chi_{t+\tau}(Z_0', \lambda) \qquad \text{for all } t \geq 0.$$

Because $\chi_t(Z_0', \lambda)$ is strictly increasing in t, this implies

$$\chi_t(Z_0', \lambda) < \chi_t(Z_0, \lambda) \qquad \text{for all } t \geq 0.$$

Thus the integrand in INV is pointwise larger under the lower initial capital stock. Therefore

$$INV(Z_0', \lambda) > INV(Z_0, \lambda).$$

Equivalently,

$$R(\lambda; Z_0', H_0) < R(\lambda; Z_0, H_0) \qquad \text{for all } \lambda \text{ whenever } Z_0' < Z_0.$$

Evaluating at the old root $\lambda^*(Z_0, H_0)$ yields

$$R(\lambda^*(Z_0, H_0); Z_0', H_0) < 0.$$

Since R is increasing in λ, the value of λ required to restore $R = 0$ must again be larger. Therefore

$$\lambda^*(Z_0', H_0) > \lambda^*(Z_0, H_0).$$

Thus λ^* is strictly decreasing in Z_0.

Combining these results with the comparative statics of long-run outcomes w.r.t. λ^*

completes the proof.□

Proof of Proposition 5. The result follows immediately because i) investment relative to GDP equals $1-\chi_t^*$, ii) rental income relative to GDP is proportional to χ_t^* when H_t is fixed, and iii) χ_t^* increases over time.□

Proof of Proposition 6. The cumulative expenditure from time 0 to time t is

$$\int_0^t \mathrm{e}^{-\rho s}\left(Y_s - C_s\right) U_{C,s}\mathrm{d}s = \int_0^t \mathrm{e}^{-\rho s}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s$$

The cumulated income from time 0 to time t is

$$\begin{aligned}
&\int_0^t \mathrm{e}^{-\rho s}\left(L_s - H_t\right) U_{L,s}\mathrm{d}s + \int_0^\infty \mathrm{e}^{-\rho s}\left(H_t - H_0\right) U_{L,s}\mathrm{d}s \\
=&U_L^*\left[\frac{1-\mathrm{e}^{-\rho t}}{\rho}\left(L^* - H_t\right) + \frac{1}{\rho}\left(H_t - H_0\right)\right] \\
=&\frac{\nu H_0^{1-\sigma}}{\rho\left(1+\eta\right)^{\sigma}}\left[\left(1-\mathrm{e}^{-\rho t}\right)\eta + \mathrm{e}^{-\rho t}\frac{H_t - H_0}{H_0}\right] \\
=&\left[\int_0^\infty \mathrm{e}^{-\rho s}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s\right]\left[\left(1-\mathrm{e}^{-\rho t}\right) + \mathrm{e}^{-\rho t}\frac{H_t - H_0}{\eta H_0}\right]
\end{aligned}$$

in which the last step uses (17). In order for the cumulated income to be larger than the cumulative expenditure (so that the government is net creditor), we need

$$\begin{aligned}
\left(1-\mathrm{e}^{-\rho t}\right) + \mathrm{e}^{-\rho t}\frac{H_t - H_0}{\eta H_0} &\geq \frac{\int_0^t \mathrm{e}^{-\rho s}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s}{\int_0^\infty \mathrm{e}^{-\rho s}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s} \\
\frac{H_t - H_0}{\eta H_0} &\geq 1 - \frac{\int_t^\infty \mathrm{e}^{-\rho(s-t)}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s}{\int_0^\infty \mathrm{e}^{-\rho s}\left(\frac{1}{\chi_s}-1\right)\mathrm{d}s}
\end{aligned}$$

the RHS of which increases in t since χ_t increases in t. When this holds as equality, SB can be achieved without borrowing by either the government or the household.□

Proof of Proposition 7. The optimality conditions w.r.t. C_t, L_t, Z_t become, for any $t \geq s$,

$$0 = U_{C,t} - \lambda^{s*}\left(Y_t + B_s^t\right) U_{CC,t} - \mu_t^{s*} \tag{C11}$$

$$0 = (1+\lambda^{s*})\, U_{L,t} + \lambda^{s*}\,(L_t - H_s)\, U_{LL,t} - \zeta_t^{s*} \tag{C12}$$

$$0 = \mu_t^{s*}\,[Y_t' - (\delta+\rho)] + \dot{\mu}_t^{s*} - \lambda^{s*} Y_t' U_{C,t} \tag{C13}$$

Define an integrating factor $m_t \equiv \mathrm{e}^{\int_0^t [Y_\tau' - (\delta+\rho)]\mathrm{d}\tau}$ and we have, using (C13)

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(m_t \frac{\mu_t^{s*}}{\lambda^{s*}}\right) &= [Y_t' - (\delta+\rho)]\, m_t \frac{\mu_t^{s*}}{\lambda^{s*}} + m_t \frac{\dot{\mu}_t^{s*}}{\lambda^{s*}} \\ &= m_t Y_t' U_{C,t}\end{aligned}$$

As its RHS is independent of s, we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left(m_t \frac{\mu_t^{s*}}{\lambda^{s*}}\right) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(m_t \frac{\mu_t^{*}}{\lambda^{*}}\right) \\ m_t \frac{\mu_t^{s*}}{\lambda^{s*}} - m_s \frac{\mu_s^{s*}}{\lambda^{s*}} &= m_t \frac{\mu_t^{*}}{\lambda^{*}} - m_s \frac{\mu_s^{*}}{\lambda^{*}}\end{aligned}$$

and thus

$$\frac{\mu_t^{s*}}{\lambda^{s*}} = \frac{\mu_t^{*}}{\lambda^{*}} + m_t^{-1} m_s \left(\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^{*}}{\lambda^{*}}\right) \tag{C14}$$

To determine the bond position, rewrite (C11) as

$$B_s^t = \frac{U_{C,t} - \mu_t^{s*}}{\lambda^{s*} U_{CC,t}} - Y_t$$

whose $s = 0$ case is $0 = \frac{U_{C,t} - \mu_t^{*}}{\lambda^{*} U_{CC,t}} - Y_t$. Taking their difference to cancel Y_t and plugging in (C14) in the last step gives

$$\begin{aligned}B_s^t &= \frac{U_{C,t} - \mu_t^{s*}}{\lambda^{s*} U_{CC,t}} - \frac{U_{C,t} - \mu_t^{*}}{\lambda^{*} U_{CC,t}} = -C_t^*\left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*}\right) + (C_t^*)^2\left(\frac{\mu_t^{s*}}{\lambda^{s*}} - \frac{\mu_t^{*}}{\lambda^{*}}\right) \\ &= -C_t^*\left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*}\right) + (C_t^*)^2\, m_t^{-1} m_s \left(\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^{*}}{\lambda^{*}}\right)\end{aligned} \tag{C15}$$

As for H_s, (C12) implies $L^* = [1 + \eta(\lambda^{s*})]\, H_s$. The logic is the same as Lemma 2.

The chosen portfolio has to satisfy the time-s implementability constraint

$$\underbrace{\int_s^\infty \mathrm{e}^{-\rho(t-s)}\,(L_t - H_s)\, U_{L,t}\mathrm{d}t}_{\equiv INC^{s*}} = \underbrace{\int_s^\infty \mathrm{e}^{-\rho(t-s)}\,(Y_t - C_t)\, U_{C,t}\mathrm{d}t}_{\equiv INV^{s*}} + \underbrace{\int_s^\infty \mathrm{e}^{-\rho(t-s)} B_s^t U_{C,t}\mathrm{d}t}_{\equiv \mathcal{B}^{s*}}$$

The land income term is, using $L_t = L^*, H_s = [1+\eta(\lambda^{s*})]^{-1} L^*$,

$$INC^{s*} = \frac{\nu U_L^* L^*}{\rho} \frac{\eta(\lambda^{s*})}{1+\eta(\lambda^{s*})} = \frac{\nu H_0^{1-\sigma}}{\rho} \frac{\eta(\lambda^{s*})}{[1+\eta(\lambda^{s*})][1+\eta(\lambda^*)]^{\sigma-1}} = \frac{\nu (L^*)^{1-\sigma}}{\rho} \frac{1+\lambda^{s*}}{\lambda^{s*}\sigma} \equiv INC(\lambda^{s*}; L^*) \tag{C16}$$

which decreases in λ^{s*} and otherwise does not depend on calendar time s.[13]

The present-sum util-unit of investment equals

$$INV^{s*} = \int_s^\infty e^{-\rho(t-s)} \left(\frac{1}{\chi_t^*} - 1 \right) dt \tag{C17}$$

which decreases in s.

The debt repayment term is, plugging in (C15),

$$\begin{aligned}
\mathcal{B}^{s*} &= \int_s^\infty e^{-\rho(t-s)} \left[-\left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + C_t^* m_t^{-1} m_s \left(\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^*}{\lambda^*} \right) \right] dt \\
&= -\frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + m_s \left(\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^*}{\lambda^*} \right) \int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt
\end{aligned}$$

Taking stock, the time-s implementability constraint is

$$INC(\lambda^{s*}; L^*) = INV^{s*} - \frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + m_s \left(\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^*}{\lambda^*} \right) \int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt$$

which determines

$$\frac{\mu_s^{s*}}{\lambda^{s*}} - \frac{\mu_s^*}{\lambda^*} = \left(m_s \int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt \right)^{-1} \left[\frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + INC(\lambda^{s*}; L^*) - INV^{s*} \right]$$

Plugging this back into (C15), we get

$$\begin{aligned}
B_s^\tau &= -C_\tau^* \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + (C_\tau^*)^2 m_\tau^{-1} \left(\int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt \right)^{-1} \left[\frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) + INC(\lambda^{s*}; L^*) - INV^{s*} \right] \\
&= C_\tau^* \left[\frac{C_\tau^* m_\tau^{-1}}{\int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt} \left(INC(\lambda^{s*}; L^*) - INV^{s*} \right) + \left(\frac{C_\tau^* m_\tau^{-1}}{\rho \int_s^\infty e^{-\rho(t-s)} C_t^* m_t^{-1} dt} - 1 \right) \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) \right]
\end{aligned}$$

[13] Note that in this analysis, we hold fixed L^* and find INC^{s*} to be decreasing in λ^{s*}, which does not contradict the finding in Section 3.2 that INC increases in λ^* when holding H_0 fixed.

The second term in the bracket averages to zero, weighted by $\mathrm{e}^{-\rho(t-s)}$, since $\rho \int_s^\infty \mathrm{e}^{-\rho(t-s)} C_t^* m_t^{-1} \mathrm{d}t$ is exactly the weighted average of $C_\tau^* m_\tau^{-1}$ across all $\tau \geq s$. As for the first term, since INV^{s*} decreases in s and $INC(\lambda^{s*}; L^*)$ decreases in λ^{s*}, there exists a threshold $\bar{\lambda}^s$ such that the first term is positive at time s if and only if $\lambda^{s*} > \bar{\lambda}^s$. The threshold $\bar{\lambda}^s$ increases over s, and in particular, its time-0 value is $\bar{\lambda}^0 = \lambda^*$.

We conclude with an observation that, for any $s > 0$ and λ^{s*}, B_s^τ / C_τ^* is either a positive constant across τ or strictly monotonic in τ. This is because

$$B_s^\tau / C_\tau^* = \frac{C_\tau^* m_\tau^{-1}}{\int_s^\infty \mathrm{e}^{-\rho(t-s)} C_t^* m_t^{-1} \mathrm{d}t} \left[\left(INC(\lambda^{s*}; L^*) - INV^{s*} \right) + \frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) \right] - \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right)$$

as a sum of a term that varies in τ and a constant. The coefficient of the first term strictly decreases in τ, since

$$\frac{\mathrm{d}\left(C_t^* / m_t \right) / \mathrm{d}t}{C_t^* / m_t} = \frac{\dot{C}_t^*}{C_t^*} - \frac{\dot{m}_t}{m_t} = -\frac{\lambda^*}{\lambda^* + \chi_t^*} \left[\chi_t^* Y_t' (Z_t^*) + \frac{\dot{\chi}_t^*}{\chi_t^*} \right] < 0$$

using (15) and $\dot{\chi}_t^* > 0$ from Proposition 3. Further, $\left(INC(\lambda^{s*}; L^*) - INV^{s*} \right) + \frac{1}{\rho} \left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right)$ is decreasing in λ^{s*} and positive when $\lambda^{s*} = \lambda^*$, since $INC(\lambda^{s*}; L^*) = INC^* = INV^* > INV^{s*}$ for $s > 0$. Thus it admits only one zero that is larger than λ^*. At that zero, the constant $-\left(\frac{1}{\lambda^{s*}} - \frac{1}{\lambda^*} \right) > 0$, suggesting that the government borrows with a "flat" (relative to consumption C_τ^*) maturity structure. For any λ^{s*} below (/above) that zero, B_s^τ / C_τ^* strictly decreases (/increases) in τ.□

Proof of Proposition B2. Proposition 7 indicates that SB can be implemented in a time-consistent way, if and only if

$$H_s \leq \frac{\sigma - 1}{\sigma} L^*$$

Proposition 6 suggests that the threshold H_s^{NB} for no-borrowing implementation increases over time. Thus in order to find an implementation that is both time-consistent and free from borrowing, we need

$$H_\infty^{NB} \leq \frac{\sigma - 1}{\sigma} L^*$$

$$H_0 + \left[1 - \frac{\frac{1}{\chi^*_\infty} - 1}{\rho \int_0^\infty \mathrm{e}^{-\rho t} \left(\frac{1}{\chi^*_t} - 1\right) \mathrm{d}t}\right] \eta\left(\lambda^*\right) H_0 \leq \frac{\sigma - 1}{\sigma} L^*$$

Using (16) and (17), we simplify it to

$$\begin{aligned}
\frac{1}{1+\eta\left(\lambda^*\right)} + \left[1 - \frac{\frac{1}{\chi^*_\infty} - 1}{\nu \left(L^*\right)^{1-\sigma} \frac{\eta(\lambda^*)}{1+\eta(\lambda^*)}}\right] \frac{\eta\left(\lambda^*\right)}{1+\eta\left(\lambda^*\right)} &\leq \frac{\sigma - 1}{\sigma} \\
-\frac{\frac{1}{\chi^*_\infty} - 1}{\nu \left(L^*\right)^{1-\sigma} \frac{\eta(\lambda^*)}{1+\eta(\lambda^*)}} \frac{\eta\left(\lambda^*\right)}{1+\eta\left(\lambda^*\right)} &\leq -\frac{1}{\sigma} \\
\frac{1}{\chi^*_\infty} - 1 &\geq \frac{\nu \left(L^*\right)^{1-\sigma}}{\sigma} \qquad \text{(C18)}
\end{aligned}$$

Note that the LHS decreases in λ^* from Proposition 4, and the RHS increases in λ^* from Lemma 2. Thus this inequality (C18) can only hold if λ^* is small.

Last, we establish a sufficient condition under which the inequality (C18) cannot hold for any $\lambda^* > 0$. We simply need (C18) to hold with weak inequality to the opposite direction under $\lambda^* = 0$, i.e.

$$\begin{aligned}
\frac{1}{\chi^{FB}_\infty} - 1 &\leq \frac{\nu}{\sigma} \\
\frac{\alpha\delta}{(1-\alpha)\delta + \rho} &\leq \frac{\nu}{\sigma}
\end{aligned}$$

□

Proof of Proposition 8. For simplicity I suppress the superscript DD. The FOCs w.r.t. C_t, L_t of (30) are

$$\begin{aligned}
0 &= U_{C,t}\Delta t + \lambda_t U_{C,t}\Delta t + \lambda_t C_t U_{CC,t}\Delta t - \lambda_t Y_t U_{CC,t}\Delta t - \mu_t \Delta t = U_{C,t}\Delta t - \lambda_t \Delta t Y_t U_{CC,t} - \mu_t \Delta t \\
0 &= \left(1 + \lambda_t\right) U_{L,t}\Delta t + \lambda_t \left(L_t - H_t\right) U_{LL,t}\Delta t - \zeta_t \Delta t
\end{aligned}$$

and the limits of $\Delta t \to 0$ are, assuming an interior solution $L_t < 1$,

$$\begin{aligned}
0 &= U_{C,t} - \lambda_t Y_t U_{CC,t} - \mu_t \\
0 &= \left(1 + \lambda_t\right) U_{L,t} + \lambda_t \left(L_t - H_t\right) U_{LL,t} - \zeta_t
\end{aligned}$$

Thus

$$\mu_t = U_{C,t} - \lambda_t Y_t U_{CC,t} \tag{C19}$$

and

$$L_t = \min\left\{1, \left[1+\eta\left(\lambda_t\right)\right] H_t\right\}$$

with excess supply function $\eta(\lambda) \equiv \left(\frac{\lambda\sigma}{1+\lambda} - 1\right)^{-1}$ which decreases in λ;

The envelope conditions from (30) are

$$\begin{aligned}
\mathcal{L}_H^{DD}\left(H_t, Z_t\right) &= -\lambda_t \left(U_{L,t}\Delta t + \mathrm{e}^{-\rho\Delta t}\mathcal{U}_{L,t+\Delta t}\right) = -\lambda_t \mathcal{P}_t \\
\mathcal{L}_Z^{DD}\left(H_t, Z_t\right) &= \mu_t \left[Y_t'\Delta t + (1-\delta\Delta t)\right] - \lambda_t Y_t' U_{C,t}\Delta t
\end{aligned}$$

The optimality conditions w.r.t. $H_{t+\Delta t}, Z_{t+\Delta t}$ of (30) are

$$\begin{aligned}
0 &= \mathrm{e}^{-\rho\Delta t}\mathcal{L}_{H,t+\Delta t}^{DD} + \lambda_t \mathrm{e}^{-\rho\Delta t}\mathcal{U}_{L,t+\Delta t} + \mathrm{e}^{-\rho\Delta t}\lambda_t \Delta H_{t+\Delta t}\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial H_{t+\Delta t}} - \xi_t \\
&= \mathrm{e}^{-\rho\Delta t}\left[\left(-\lambda_{t+\Delta t} + \lambda_t\right)\mathcal{U}_{L,t+\Delta t} + \lambda_t \Delta H_{t+\Delta t}\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial H_{t+\Delta t}}\right] - \xi_t
\end{aligned} \tag{C20}$$

$$\begin{aligned}
0 &= \mathrm{e}^{-\rho\Delta t}\mathcal{L}_{Z,t+\Delta t}^{DD} - \mu_t + \mathrm{e}^{-\rho\Delta t}\lambda_t \Delta H_{t+\Delta t}\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial Z_{t+\Delta t}} \\
&= \mathrm{e}^{-\rho\Delta t}\left\{\mu_{t+\Delta t}\left[Y_{t+\Delta t}'\Delta t + (1-\delta\Delta t)\right] - \lambda_{t+\Delta t} Y_{t+\Delta t}' U_{C,t+\Delta t}\Delta t + \lambda_t \Delta H_{t+\Delta t}\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial Z_{t+\Delta t}}\right\} - \mu_t
\end{aligned} \tag{C21}$$

In simplifying last two expressions, we have plugged in the envelope conditions.

When $H_{t+\Delta t} < 1$ and thus $\xi_t = 0$, the limit of $\Delta t \to 0$ of $\mathrm{e}^{\rho\Delta t}\times$(C20) is

$$0 = -\frac{\dot{\lambda}_t}{\lambda_t}\mathcal{P}_t + \dot{H}_t\left(\frac{\partial \mathcal{P}}{\partial H}\right)_t$$

which implies

$$\dot{H}_t = \frac{\frac{\dot{\lambda}_t}{\lambda_t}\mathcal{P}_t}{\left(\frac{\partial \mathcal{P}}{\partial H}\right)_t} \tag{C22}$$

The limit of $\Delta t \to 0$ of $\mathrm{e}^{\rho\Delta t}\times$(C21) is

$$\begin{aligned}
0 &= \mu_{t+\Delta t}\left[Y_{t+\Delta t}'\Delta t + (1-\delta\Delta t)\right] - \lambda_{t+\Delta t} Y_{t+\Delta t}' U_{C,t+\Delta t}\Delta t + \lambda_t \Delta H_{t+\Delta t}\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial Z_{t+\Delta t}} - \mu_t \mathrm{e}^{\rho\Delta t} \\
&= \mu_{t+\Delta t}\Delta t\left(Y_{t+\Delta t}' - \delta\right) + \dot{\mu}_t\Delta t - \rho\mu_t\Delta t - \lambda_{t+\Delta t} Y_{t+\Delta t}' U_{C,t+\Delta t}\Delta t + \lambda_t \dot{H}_t \Delta t\frac{\partial \mathcal{U}_{L,t+\Delta t}}{\partial Z_{t+\Delta t}}
\end{aligned}$$

$$0 = \mu_t \left(Y'_t - \rho - \delta\right) + \dot{\mu}_t - \lambda_t Y'_t U_{C,t} + \lambda_t \dot{H}_t \left(\frac{\partial \mathcal{P}}{\partial Z}\right)_t$$

Plugging in (C19) and then (C22), we get

$$\begin{aligned}
0 &= \left(U_{C,t} - \lambda_t Y_t U_{CC,t}\right)\left(Y'_t - \rho - \delta\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(U_{C,t} - \lambda_t Y_t U_{CC,t}\right) - \lambda_t Y'_t U_{C,t} + \lambda_t \dot{H}_t \left(\frac{\partial \mathcal{P}}{\partial Z}\right)_t \\
&= C_t^{-1}\left(1 + \lambda_t \chi_t^{-1}\right)\left(Y'_t - \rho - \delta\right) - C_t^{-2}\dot{C}_t\left(1 + \lambda_t \chi_t^{-1}\right) + C_t^{-1}\dot{\lambda}_t \chi_t^{-1} - C_t^{-1}\lambda_t \chi_t^{-2}\dot{\chi}_t - \lambda_t Y'_t U_{C,t} + \lambda_t \dot{H}_t \left(\frac{\partial \mathcal{P}}{\partial Z}\right)_t \\
\frac{\dot{C}_t}{C_t} &= \left[Y'_t\left(Z_t\right) - \left(\delta + \rho\right)\right] - \frac{\lambda_t}{\lambda_t + \chi_t}\left[\chi_t Y'_t\left(Z_t\right) + \frac{\dot{\chi}_t}{\chi_t} - \frac{\dot{\lambda}_t}{\lambda_t} - \chi_t C_t \dot{H}_t \left(\frac{\partial \mathcal{P}}{\partial Z}\right)_t\right] \\
&= \left[Y'_t\left(Z_t\right) - \left(\delta + \rho\right)\right] - \frac{\lambda_t}{\lambda_t + \chi_t}\left[\chi_t Y'_t\left(Z_t\right) + \frac{\dot{\chi}_t}{\chi_t} - \frac{\dot{\lambda}_t}{\lambda_t} - \chi_t C_t \frac{\mathcal{P}_t}{\left(\partial \mathcal{P}/\partial H\right)_t}\frac{\dot{\lambda}_t}{\lambda_t}\left(\frac{\partial \mathcal{P}}{\partial Z}\right)_t\right] \\
&= \left[Y'_t\left(Z_t\right) - \left(\delta + \rho\right)\right] - \frac{\lambda_t}{\lambda_t + \chi_t}\left[\chi_t Y'_t\left(Z_t\right) + \frac{\dot{\chi}_t}{\chi_t} - \left(1 + \chi_t C_t \frac{\mathcal{P}_t \left(\partial \mathcal{P}/\partial Z\right)_t}{\left(\partial \mathcal{P}/\partial H\right)_t}\right)\frac{\dot{\lambda}_t}{\lambda_t}\right]
\end{aligned}$$

Last, the limit under $\Delta t \to 0$ of the time-t implementability condition is

$$\begin{aligned}
0 &= \left(C_t - Y_t\left(Z_t\right)\right) U_{C,t}\Delta t + \left(L_t - H_t\right) U_{L,t}\Delta t + \mathrm{e}^{-\rho \Delta t}\Delta H_{t+\Delta t}\boldsymbol{\mathcal{U}}_{L,t+\Delta t} \\
0 &= \left(C_t - Y_t\left(Z_t\right)\right) U_{C,t} + \left(L_t - H_t\right) U_{L,t} + \dot{H}_t \mathcal{P}_t
\end{aligned}$$

which simplifies to

$$\nu L_t^{-\sigma}\left(L_t - H_t\right) + \dot{H}_t \mathcal{P}_t = \frac{1}{\chi_t} - 1$$

□

Proof of Proposition 9. For simplicity I suppress the superscript DD when there is no ambiguity.

Steady state. If the steady state $L_\infty < 1$, then we must have $\lambda_\infty > 0$ and $L_\infty = \left[1 + \eta\left(\lambda_\infty\right)\right] H_\infty$ from (33). The fiscal budget (35) implies that

$$\nu L_\infty^{1-\sigma}\frac{\eta\left(\lambda_\infty\right)}{1 + \eta\left(\lambda_\infty\right)} = \frac{1}{\chi_\infty} - 1$$

which implies

$$\nu L_\infty^{1-\sigma} = \frac{1+\eta}{\eta}\left(\chi_\infty^{-1} - 1\right) = \frac{\lambda_\infty \sigma}{\lambda_\infty + 1}\left(\chi_\infty^{-1} - 1\right) \tag{C23}$$

We can plug in (18) which expresses χ_∞ in terms of λ_∞ as

$$\chi_\infty(\lambda_\infty) = \frac{(1-\lambda_\infty) + \sqrt{(1+\lambda_\infty)^2 + \frac{4\delta\alpha}{\delta(1-\alpha)+\rho}\lambda_\infty}}{2}\frac{\delta(1-\alpha)+\rho}{\delta+\rho}$$

It applies in the DD case too since it only involves the consumption growth ODE and resource constraint, whose steady-state form is the same as in SB. It implies that $\chi^{-1} > 1$ if $\lambda = 0$ and $\chi^{-1} \to 1$ if $\lambda \to \infty$. Hence the RHS of (C23) obtains 0 when $\lambda = 0$ or $\lambda \to \infty$.

Next we establish that $f(\lambda) = \frac{\lambda}{\lambda+1}\left((\chi(\lambda))^{-1} - 1\right)$ is strictly increasing on $(0,1)$ and strictly decreasing on $(1,\infty)$ and hence obtains maximum at 1. Let $\kappa = \frac{4\delta\alpha}{\delta(1-\alpha)+\rho}$, $m = \frac{\delta(1-\alpha)+\rho}{\delta+\rho}$, $D(\lambda) = (\lambda+1)^2 + \kappa\lambda$ and we have

$$\chi(\lambda) = m\frac{1-\lambda+\sqrt{D(\lambda)}}{2}$$
$$(\chi(\lambda))^{-1} = \frac{2m^{-1}}{1-\lambda+\sqrt{D(\lambda)}} = \frac{2(1+\kappa/4)}{1-\lambda+\sqrt{D(\lambda)}}$$

and thus

$$f(\lambda) = \frac{\lambda}{\lambda+1}\left(\frac{2(1+\kappa/4)}{\sqrt{D(\lambda)}+1-\lambda} - 1\right) = \frac{\lambda(\kappa+2\lambda+2-2\sqrt{D(\lambda)})}{2(\lambda+1)(\sqrt{D(\lambda)}+1-\lambda)}$$

Hence,

$$f'(\lambda) = -\frac{\kappa(\lambda-1)}{4(\lambda+1)^2\sqrt{D(\lambda)}}$$

which implies that it is increasing on $(0,1)$, decreasing on $(1,\infty)$ and hence obtains maximum at 1 of

$$f(1) = \frac{1}{2}\left((\chi(1))^{-1} - 1\right) = \frac{1}{2}\left(\sqrt{\frac{\delta+\rho}{\delta(1-\alpha)+\rho}} - 1\right)$$

Therefore, according to (C23), the minimum L_∞ satisfies

$$L_\infty \geq \underline{\mathrm{L}} = \left[\frac{\sigma}{2\nu}\left(\sqrt{\frac{\delta+\rho}{\delta(1-\alpha)+\rho}} - 1\right)\right]^{1/(1-\sigma)}$$

If this is larger than 1, there is no interior steady state. That is

$$\frac{\delta+\rho}{\delta(1-\alpha)+\rho} < \left(1+\frac{2\nu}{\sigma}\right)^2$$
$$\frac{\delta\alpha}{\delta(1-\alpha)+\rho} < 4\frac{\nu}{\sigma}\left(\frac{\nu}{\sigma}+1\right)$$

Local convergence. Around the steady state of $L_\infty = 1$ which is independent of (Z, H), we have $\mathcal{P} = \frac{\nu}{\rho}$ locally and hence $\frac{\partial\mathcal{P}}{\partial Z} = \frac{\partial\mathcal{P}}{\partial H} = 0$ locally.

Then (34) implies $\dot{\lambda}_t = 0$ and thus $\lambda_t = \lambda$. Thus (32) simplifies to

$$\frac{\dot{C}_t}{C_t} = \left[Y_t'(Z_t) - (\delta+\rho)\right] - \frac{\lambda}{\lambda+\chi_t}\left[\chi_t Y_t'(Z_t) + \frac{\dot{\chi}_t}{\chi_t}\right]$$

Together with the resource constraint (2) and $\chi_t = C_t/Y_t$, these form the same ODE system as in Proposition 3, though λ may be different. The proof of Proposition 3 extends here and implies that $\dot{\chi}_t > 0$.

Further, (35) implies that

$$\nu(1-H_t) + \dot{H}_t\mathcal{P} = \frac{1}{\chi_t} - 1$$

which determines

$$H_t = 1 + \nu^{-1}\left(1 - a\int_t^\infty e^{-a(s-t)}\frac{1}{\chi_s}ds\right)$$

with $a = \nu\mathcal{P}^{-1}$. As χ_t goes up over time, $1/\chi_t$ decreases and thus the integral term which is a discounted future average decreases. Therefore, H_t rises over time to converge to $H_\infty = 1 - \nu^{-1}\left(\chi_\infty^{-1} - 1\right)$.□

Proof of Proposition 10. Though the Proposition is stated for a generic τ, note that we are studying a Markov Perfect Equilibrium and thus without loss of generality we can start our analysis at time τ, i.e. relabeling $t-\tau$ as t and focusing on later periods.

For simplicity I suppress the superscript DD. Assume that $\dot{Z}_t > 0$, $\left(\frac{\partial\mathcal{P}}{\partial Z}\right)_t < 0$ and $\left(\frac{\partial\mathcal{P}}{\partial H}\right)_t < 0$ for all t. Assume that $L_t \in \left(\frac{\sigma}{\sigma-1}H_s, 1\right)$ for all $t \geq s \geq 0$, and thus (33) takes the interior solution $L_t = [1+\eta(\lambda_t)]H_t$. Further, assume that χ_t is strictly monotonic and λ_t is twice differentiable.

Step 1: If $\dot{\lambda}$ has the same sign as $\dot{\chi}$ at some time s, they must have the same sign for all $t \geq s$. For any τ such that $\dot{\lambda}_\tau = 0$, plug (33, 34) into (35) and evaluate its time derivative at τ,

$$\frac{(\mathcal{U}_{L,\tau})^2}{(\partial\mathcal{P}/\partial H)_\tau}\frac{\ddot{\lambda}_\tau}{\lambda_\tau} = -\frac{\dot{\chi}_\tau}{(\chi_\tau)^2}$$

As $\left(\frac{\partial\mathcal{P}}{\partial H}\right)_\tau < 0$, $\ddot{\lambda}_\tau$ must have the same sign as $\dot{\chi}_\tau$. Thus $\dot{\lambda}$ crosses zero at most once, and after that point, $\dot{\lambda}$ must have the same sign as $\dot{\chi}$.

Step 2: $\dot{\lambda}_t$ must have the opposite sign from $\dot{\chi}_t$ for all t. From step 1, if $\dot{\lambda}_t$ ever has the same sign as $\dot{\chi}_t$ at some point, there must exists $s \geq 0$ such that $\dot{\lambda}_t$ has the same sign as $\dot{\chi}_t$ for all $t \geq s$. I show that this cannot happen. Integrate $\mathrm{e}^{-\rho(s-t)}\times$(35) from such a time s to infinity and get the LHS as

$$\begin{aligned}
INC^s &\equiv \int_s^\infty \mathrm{e}^{-\rho(t-s)}\left[U_{L,t}(L_t - H_t) + \mathcal{P}_t\dot{H}_t\right]\mathrm{d}t \\
&= \int_s^\infty \mathrm{e}^{-\rho(t-s)}U_{L,t}(L_t - H_t)\,\mathrm{d}t + \int_s^\infty \mathrm{e}^{-\rho(t-s)}\mathcal{P}_t\mathrm{d}H_t \\
&= \int_s^\infty \mathrm{e}^{-\rho(t-s)}U_{L,t}(L_t - H_t)\,\mathrm{d}t - \mathcal{U}_{L,s}H_s - \int_s^\infty H_t\mathrm{d}\left(\mathrm{e}^{-\rho(t-s)}\mathcal{P}_t\right) \\
&= \int_s^\infty \mathrm{e}^{-\rho(t-s)}U_{L,t}(L_t - H_s)\,\mathrm{d}t
\end{aligned}$$

The intuition for this transformation is that future land sale is fairly priced too, the same as why H_t does not appear in the time-0 implementability condition (12) under SB allocation. Now we plug in the optimal land supply (33) and write

$$INC^s = \int_s^\infty \mathrm{e}^{-\rho(t-s)}\iota_{t,s}\mathrm{d}t$$

with $\iota_{t,s} = \nu\left[(1+\eta_t)^{1-\sigma}H_t^{1-\sigma} - (1+\eta_t)^{-\sigma}H_t^{-\sigma}H_s\right], \eta_t = \eta(\lambda_t)$. That implies that ρINC^s is a weighted average of $\iota_{t,s}$ over $t \geq s$. If $\iota_{t,s}$ is strictly decreasing (/increasing) for all $t \geq s$, we would have $\iota_{s,s}$ to be strictly larger (/smaller) than ρINC^s. We take the derivative of $\iota_{t,s}$ w.r.t. t to get

$$\frac{\mathrm{d}\iota_{t,s}}{\mathrm{d}t} = \nu(1+\eta_t)^{-\sigma}H_t^{-\sigma}\left[(1-\sigma)\eta_t'\dot{\lambda}_t H_t + (1-\sigma)(1+\eta_t)\dot{H}_t + \sigma\frac{\eta_t'\dot{\lambda}_t}{1+\eta_t}H_s + \sigma\frac{\dot{H}_t}{H_t}H_s\right]$$

$$= -\nu\left(\sigma-1\right)\left(1+\eta_t\right)^{-\sigma}H_t^{-\sigma-1}\left[\left(1+\eta_t\right)H_t - \frac{\sigma}{\sigma-1}H_s\right]\left(\dot{H}_t + \frac{\eta_t'\dot{\lambda}_t}{1+\eta_t}H_t\right)$$
$$= -\nu\left(\sigma-1\right)\left(1+\eta_t\right)^{-\sigma}H_t^{-\sigma-1}\left(L_t - \frac{\sigma}{\sigma-1}H_s\right)\left(\frac{\mathcal{P}_t}{\left(\partial\mathcal{P}/\partial H\right)_t}\frac{1}{\lambda_t} + \frac{\eta_t'}{1+\eta_t}H_t\right)\dot{\lambda}_t$$

of which the last step uses (33) and (34). Since $L_t > \frac{\sigma}{\sigma-1}H_s$, $\left(\partial\mathcal{P}/\partial H\right)_t < 0$, $\eta' < 0$, and $\sigma > 1$, we observe that $\frac{\mathrm{d}\iota_{t,s}}{\mathrm{d}t}$ has the same sign as $\dot{\lambda}_t$. Thus if λ_t is strictly decreasing (/increasing) for all $t \geq s$, we would have $\iota_{s,s}$ to be strictly larger (/smaller) than ρINC^s.

Note that $INV^s \equiv \int_s^\infty \mathrm{e}^{-\rho(t-s)}\left(\frac{1}{\chi_t}-1\right)\mathrm{d}t$ also implies that ρINV^s is a weighted average of $\frac{1}{\chi_t}-1$ for all $t \geq s$. Thus if χ_t is strictly increasing (/decreasing), then $\frac{1}{\chi_s}-1$ is strictly larger (/smaller) than ρINV^s.

As $INC^s = INV^s$ holds for all s, we take the derivative w.r.t. s and get

$$\rho INC^s - \iota_{s,s} - \mathcal{U}_{L,s}\dot{H}_s = \rho INV^s - \left(\frac{1}{\chi_s}-1\right)$$

Now supposing that $\dot{\chi}_t > 0$ for all t, we would have $\rho INV^s - \left(\frac{1}{\chi_s}-1\right) < 0$ and thus $\rho INC^s - \iota_{s,s} - \mathcal{U}_{L,s}\dot{H}_s < 0$. If $\dot{\lambda}_t \geq 0$ for all t, we would have $\dot{H}_s \leq 0$ from (34) and $\iota_{s,s} \leq \rho INC^s$, which together imply $\rho INC^s - \iota_{s,s} - \mathcal{U}_{L,s}\dot{H}_s \leq 0$ — a contradiction! Thus if $\dot{\chi}_t > 0$ for all t, we must have $\dot{\lambda}_t < 0$ for all t. Similarly, we can prove that if $\dot{\chi}_t < 0$ for all t, we must have $\dot{\lambda}_t > 0$ for all t.

Step 3: $\dot{\chi}_t < 0$ for all t. Here I further suppress the time subscript when there is no ambiguity. Similar to the proof of Proposition 3, from (32), we have

$$g^\chi = Y'\frac{\chi^2}{\chi+\lambda} - \left[\left(1-\alpha\right)\delta+\rho\right] - \frac{\lambda}{\chi+\lambda}g^\chi + \frac{\lambda}{\chi+\lambda}\left(1+\chi C\frac{\mathcal{P}\left(\partial\mathcal{P}/\partial Z\right)}{\left(\partial\mathcal{P}/\partial H\right)}\right)g^\lambda$$

with $g^\chi \equiv \frac{\dot{\chi}}{\chi}$, $g^\lambda \equiv \frac{\dot{\lambda}}{\lambda}$. Let $K_t \equiv \chi_t C_t \frac{\mathcal{P}_t\left(\partial\mathcal{P}/\partial Z\right)_t}{\left(\partial\mathcal{P}/\partial H\right)_t} > 0$ and $\gamma \equiv \frac{\chi+2\lambda}{\chi+\lambda}g^\chi - \frac{\lambda}{\chi+\lambda}\left(1+K\right)g^\lambda$ which has the same sign as g^χ. We have

$$\gamma = Y'\frac{\chi^2}{\chi+\lambda} - \left[\left(1-\alpha\right)\delta+\rho\right]$$
$$\dot{\gamma} = Y''\dot{Z}\frac{\chi^2}{\chi+\lambda} + Y'\frac{\left(\chi+2\lambda\right)\chi^2}{\left(\chi+\lambda\right)^2}g^\chi - Y'\frac{\lambda\chi^2}{\left(\chi+\lambda\right)^2}g^\lambda$$

$$= Y''\dot{Z}\frac{\chi^2}{\chi+\lambda} + Y'\frac{(\chi+2\lambda)\,\chi^2}{(\chi+\lambda)^2}g^\chi + Y'\frac{\chi^2}{\chi+\lambda}\frac{\gamma - \frac{\chi+2\lambda}{\chi+\lambda}g^\chi}{1+K}$$
$$= Y''\dot{Z}\frac{\chi^2}{\chi+\lambda} + Y'\frac{\chi^2}{\chi+\lambda}\frac{\gamma}{1+K} + Y'\frac{(\chi+2\lambda)\,\chi^2}{(\chi+\lambda)^2}g^\chi\frac{K}{1+K}$$

Next I show that $g^\chi < 0$ cannot hold. Suppose for contradiction that it holds, then $\gamma < 0$ and

$$\dot{\gamma} < Y''\dot{Z}\frac{\chi^2}{\chi+\lambda} + Y'\frac{\chi^2}{\chi+\lambda}\frac{\gamma}{1+K}$$

whose RHS is less than zero too. Thus γ will keep decreasing, which contradicts convergence. Therefore, we conclude that $\dot{\chi} > 0$. Consequently, from step 2, $\dot{\lambda} < 0$. (34) suggests that $\dot{H} > 0$. As H increases over time and $\eta(\lambda)$ decreases in λ and thus increases in time, (33) implies that $\dot{L} > 0$.□

Proof of Proposition 11. The implementability constraint is

$$0 = \int_s^\infty e^{-\rho(t-s)}\left[(C_t - Y_t)\,U_{C,t} + \left(L_t - H_0 - F_s^t - g_s\Gamma(Y_t)\right)U_{L,t}\right]dt, \tag{C24}$$

I construct the Lagrangian as

$$\mathcal{L}^{s*} = \max\int_s^\infty e^{-\rho(t-s)}U(C_t,L_t)\,dt + \int_s^\infty e^{-\rho(t-s)}\left\{\mu_t^{s*}\left[Y(Z_t) - C_t - (\delta+\rho)Z_t\right] + \dot{\mu}_t^{s*}Z_t\right\}dt + \mu_s^{s*}Z_s$$
$$+ \int_s^\infty e^{-\rho(t-s)}\zeta_t^{s*}(1-L_t)\,dt + \lambda^{s*}\int_s^\infty e^{-\rho(t-s)}\left[(C_t - Y(Z_t) - T_t)\,U_{C,t} + \left(L_t - F_s^t - g_s\Gamma(Z_t)\right)U_{L,t}\right]dt \tag{C25}$$

The optimality conditions (C2-C4) become, for any $t \geq s$,

$$0 = U_{C,t} - \lambda^{s*}Y_tU_{CC,t} - \mu_t^{s*} \tag{C26}$$
$$0 = (1+\lambda^{s*})\,U_{L,t} + \lambda^{s*}\left(L_t - H_0 - F_s^t - g_s\Gamma(Y_t)\right)U_{LL,t} - \zeta_t^{s*} \tag{C27}$$
$$0 = \mu_t^{s*}\left[Y_t' - (\delta+\rho)\right] + \dot{\mu}_t^{s*} - \lambda^{s*}\left(Y_t'U_{C,t} + g_s\Gamma_t'U_{L,t}\right) \tag{C28}$$

We need (C27) and (C24) both to hold, which means

$$H_0 + F_s^t + g_s\Gamma(Z_t) = \frac{L^*}{1+\eta(\lambda^{s*})} \tag{C29}$$

and

$$INC\left(\lambda^{s*};L^*\right)=INV^{s*}$$

with $INC\left(\lambda^{s*};L^*\right)\equiv\frac{\nu(L^*)^{1-\sigma}}{\rho}\frac{1+\lambda^{s*}}{\lambda^{s*}\sigma}$. This latter equation pins down $\lambda^{s*}>\lambda^*$ and the former one determines F_s^t for any $g_s\Gamma\left(Z_t\right)$. Then we need to find $g_s\Gamma\left(Z_t\right)$ that satisfies (C26) and (C28).

Define an integrating factor $m_t\equiv e^{\int_0^t\left[Y_\tau'-(\delta+\rho)\right]d\tau}$ and we have, using (C28)

$$\begin{aligned}\frac{d}{dt}\left(m_t\frac{\mu_t^{s*}}{\lambda^{s*}}\right)&=\left[Y_t'-(\delta+\rho)\right]m_t\frac{\mu_t^{s*}}{\lambda^{s*}}+m_t\frac{\dot{\mu}_t^{s*}}{\lambda^{s*}}\\&=m_t\left(Y_t'U_{C,t}+g_s\Gamma_t'U_{L,t}\right)\end{aligned}$$

for all s. Take the difference between a generic s and $s=0$, since $m_tY_t'U_{C,t}$ is independent of s

$$\begin{aligned}\frac{d}{dt}\left(m_t\frac{\mu_t^{s*}}{\lambda^{s*}}\right)-\frac{d}{dt}\left(m_t\frac{\mu_t^{*}}{\lambda^{*}}\right)&=m_tg_s\Gamma_t'U_{L,t}\\m_t\frac{\mu_t^{s*}}{\lambda^{s*}}-m_t\frac{\mu_t^{*}}{\lambda^{*}}&=m_s\frac{\mu_s^{s*}}{\lambda^{s*}}-m_s\frac{\mu_s^{*}}{\lambda^{*}}+g_s\int_s^t m_\tau\Gamma_\tau'U_{L,\tau}d\tau\end{aligned}$$

Rewrite (C26) as

$$0=\frac{U_{C,t}-\mu_t^{s*}}{\lambda^{s*}U_{CC,t}}-Y_t$$

and take the difference between a generic s and $s=0$ to get

$$\begin{aligned}0&=\frac{U_{C,t}-\mu_t^{s*}}{\lambda^{s*}U_{CC,t}}-\frac{U_{C,t}-\mu_t^{*}}{\lambda^{*}U_{CC,t}}=-C_t^*\left(\frac{1}{\lambda^{s*}}-\frac{1}{\lambda^*}\right)+\left(C_t^*\right)^2\left(\frac{\mu_t^{s*}}{\lambda^{s*}}-\frac{\mu_t^{*}}{\lambda^{*}}\right)\\&=-C_t^*\left(\frac{1}{\lambda^{s*}}-\frac{1}{\lambda^*}\right)+\left(C_t^*\right)^2m_t^{-1}\left[m_s\left(\frac{\mu_s^{s*}}{\lambda^{s*}}-\frac{\mu_s^{*}}{\lambda^{*}}\right)+g_s\int_s^t m_\tau\Gamma_\tau'U_{L,\tau}d\tau\right]\\0&=-\frac{m_t}{C_t^*}\left(\frac{1}{\lambda^{s*}}-\frac{1}{\lambda^*}\right)+m_s\left(\frac{\mu_s^{s*}}{\lambda^{s*}}-\frac{\mu_s^{*}}{\lambda^{*}}\right)+g_s\int_s^t m_\tau\Gamma_\tau'U_{L,\tau}d\tau\end{aligned}\tag{C30}$$

Evaluate (C30) at $t=s$

$$0=-\frac{m_s}{C_s^*}\left(\frac{1}{\lambda^{s*}}-\frac{1}{\lambda^*}\right)+m_s\left(\frac{\mu_s^{s*}}{\lambda^{s*}}-\frac{\mu_s^{*}}{\lambda^{*}}\right)$$

which determines μ_s^{s*} with $\frac{\mu_s^{s*}}{\lambda^{s*}}-\frac{\mu_s^{*}}{\lambda^{*}}<0$ since $\lambda^{s*}>\lambda^*$.

Take the derivative of (C30) w.r.t. t

$$0=\frac{m_t}{C_t^*}\left(\frac{1}{\lambda^{s*}}-\frac{1}{\lambda^*}\right)\left(\frac{\dot{C}_t^*}{C_t^*}-\frac{\dot{m}_t}{m_t}\right)+g_s m_t\Gamma_t' U_{L,t}$$

In order for this to hold, we need

$$m_t\Gamma_t' U_{L,t}=k\frac{m_t}{C_t^*}\left(\frac{\dot{C}_t^*}{C_t^*}-\frac{\dot{m}_t}{m_t}\right)$$

holds with a positive constant k. Then

$$g_s=k^{-1}\left(\frac{1}{\lambda^*}-\frac{1}{\lambda^{s*}}\right)$$

We need

$$\begin{aligned}\Gamma_t' U_{L,t}&=k\frac{1}{C_t^*}\left(\frac{\dot{C}_t^*}{C_t^*}-\left[Y'\left(Z_t^*\right)-\left(\delta+\rho\right)\right]\right)\\&=-k\frac{\lambda^*}{\lambda^*+\chi_t^*}\left(\chi_t^* Y'\left(Z_t^*\right)+\frac{\dot{\chi}_t^*}{\chi_t^*}\right)&\text{(C31)}\\&=-k\frac{\lambda^*}{\lambda^*+\chi_t^*}\left(\chi_t^* Y'\left(Z_t^*\right)+Y'\left(Z_t^*\right)\frac{\left(\chi_t^*\right)^2}{\chi_t^*+2\lambda^*}-\frac{\chi_t^*+\lambda^*}{\chi_t^*+2\lambda^*}\left[\left(1-\alpha\right)\delta+\rho\right]\right)\\&=-k\frac{\lambda^*}{\lambda^*+2\chi_t^*}\left[2\chi_t^* Y'\left(Z_t^*\right)-\left[\left(1-\alpha\right)\delta+\rho\right]\right]\\\Gamma_t'&=-k\nu^{-1}\left(L^*\right)^{\sigma}\frac{\lambda^*}{\lambda^*+2\chi_t^*}\left[2\chi_t^* Y'\left(Z_t^*\right)-\left[\left(1-\alpha\right)\delta+\rho\right]\right]&\text{(C32)}\end{aligned}$$

in which we first use (15) and then plug in (C7). (C31) implies that $\Gamma_t'<0$ — the more output, the less land payment. (C29) then pins down F_s^t.□

Proof of Proposition B1. The household lifetime budget constraint is

$$\begin{aligned}0&=\int_0^\infty Q_t\left[C_t-\left(1-\tau_y\right)Y_t\left(Z_t\right)+P_t\dot{H}_t+\tau_p P_t H_t+D_t\left(L_t-H_t\right)-T_t\right]\mathrm{d}t\\&=\int_0^\infty\left\{Q_t\left[C_t-\left(1-\tau_y\right)Y_t\left(Z_t\right)+D_t\left(L_t-H_t\right)+\tau_p P_t H_t-T_t\right]-H_t\frac{\mathrm{d}\left(P_tQ_t\right)}{\mathrm{d}t}\right\}\mathrm{d}t-P_0H_0\end{aligned}$$

Assume that the land supply is fixed at L, which nests the case of first best with $L=1$.

Their consumption optimization implies

$$
\begin{aligned}
Q_t &= \mathrm{e}^{-\rho t} U_{C,0}^{-1} U_{C,t} \\
D_t &= U_{C,t}^{-1} U_L
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{\mathrm{d}\left(P_t Q_t\right)}{\mathrm{d}t} &= Q_t \left(\tau_p P_t - D_t\right) \\
P_t &= Q_t^{-1} \int_t^{\infty} \mathrm{e}^{-\tau_p (s-t)} Q_s D_s \mathrm{d}s = \frac{1}{\tau_p + \rho} U_{C,t}^{-1} U_L
\end{aligned}
$$

Since the household is indifferent about H_t, we can evaluate the present value of land tax income plus the land sale and rent income by assuming $H_t = H_0$ and get

$$
\begin{aligned}
\int_0^{\infty} Q_t \left[P_t \dot{H}_t + \tau_p P_t H_t + D_t \left(L_t - H_t\right)\right] \mathrm{d}t &= \int_0^{\infty} Q_t \left[\tau_p P_t H_0 + D_t \left(L - H_0\right)\right] \mathrm{d}t \\
&= U_{C,0}^{-1} U_L \int_0^{\infty} \mathrm{e}^{-\rho t} \left(\frac{\tau_p}{\tau_p + \rho} H_0 + L - H_0\right) \mathrm{d}t \\
&= U_{C,0}^{-1} \frac{\nu}{\rho} L^{-\sigma} \left(L - \frac{\rho}{\tau_p + \rho} H_0\right)
\end{aligned}
$$

which obtains $U_{C,0}^{-1} \frac{\nu}{\rho} \left(1 - \frac{\rho}{\tau_p + \rho} H_0\right)$ if $L = 1$.

As analyzed before, the present value of investment minus income tax is

$$
\int_0^{\infty} Q_t \left[\left(1 - \tau_y\right) Y_t \left(Z_t\right) - C_t\right] \mathrm{d}t = U_{C,0}^{-1} \int_0^{\infty} \mathrm{e}^{-\rho t} \left(\frac{1}{\chi_t} - 1\right) \mathrm{d}t
$$

Therefore, if

$$
\frac{\nu}{\rho} \left(1 - \frac{\rho}{\tau_p + \rho} H_0\right) < \int_0^{\infty} \mathrm{e}^{-\rho t} \left(\frac{1 - \tau_y}{\chi_t^{FB}} - 1\right) \mathrm{d}t
$$

first best cannot be implemented.□